Project Requirements in IT 311 - Analytics Modeling

2nd Semester 2024-2025

Project Title	**Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records**				
Team Members	Arrange your names based on the percentage of members' contribution. Student 1 with the most contribution.		**Score**		
			Individual Presentation (100)	**Group Grade**	
	Last name, First name	*Role in the project development*		*Building Analytics Model (200)*	*System Development (150)*
1	**Chico, Ken**	**Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Live Presenter, Quality Checker**			
2	**De Jesus, Princess Anne**	**Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Video Specialist, Quality Checker**			
3	**Garces, Grace**	**Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Video Specialist**			
4	**Lachica, Crystel**	**Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Video Specialist**			
5	**Macabenta, Laarnie**	**Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Video Specialist, Quality Checker**			
6	**Mendoza, Carl Andrei**	**Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Live Presenter, Project Leader**			
			Total Group Grade:		

Individual Presentation Rubrics

Criteria	**Weight**
The screencast for the system presentation demonstrates the complete process for building an analytics model.	15
The screencast for the system presentation demonstrates how the developed system is used for predicting a target label.	15
The screencast for visual analytics shows all the possible insights that can be derived from the dataset.	20
The team answers all the questions correctly and promptly.	20
Video presentations show creativity.	15
The team is well-prepared and confident throughout the presentation.	10
The team presented the output within the allotted time for their group.	5
Total	**100**

BULACAN STATE UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
City of Malolos, Bulacan

IT 311: ANALYTICS MODELING

Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records

Final Project in IT311

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1. Project Rationale

1.1. Context and Background of the Dataset

Crime occurs as a result of complex decision-making processes, influenced by both an individual's predisposition and their exposure to high-risk environments (Ceccato & Nalla, 2020). Traditional crime analysis relies heavily on manual methods, which are time-consuming and can hinder proactive law enforcement efforts. However, with advancements in data mining and machine learning, predictive modeling now allows law enforcement to anticipate criminal activity based on historical trends.

This study utilizes crime data from the Calumpit Municipal Police Station (CMPS) of the Philippine National Police, covering incidents from 2013-2025. The raw dataset includes 83 attributes that contain essential details such as time, location, and other relevant case information, offering valuable insights for law enforcement applications. Machine learning-driven crime analysis enhances investigative processes, and optimizes resource allocation, ensuring that cases with a higher likelihood of resolution receive appropriate attention.

By leveraging predictive analytics, law enforcement can improve crime prevention strategies and make data-driven decisions regarding case management while victims can use predictive insights to assess if legal action is necessary, ensuring that situations with a better likelihood of resolution receive prompt attention. Ultimately, the goal is to provide law enforcement and victims with structured insights into whether a case—based on available details—is likely to be solved.

1.2. Importance and Value of the Project

The importance of this project is to give crime victims a clearer sense of whether their case is likely to be solved. Many victims, especially those without suspect information or strong evidence, often feel uncertain about whether pursuing justice will lead to meaningful outcomes. The predictive tool uses historical data to predict case outcomes. This helps victims understand what to expect and make informed decisions.

Implementing a predictive analytic model offers a significant competitive advantage by enhancing operational efficiency, improving decision-making, and strengthening the strategic capabilities of the justice system, ultimately fostering greater transparency and confidence among victims as they navigate the legal process (Walczak, 2021). Through this mechanism, law enforcement can orient itself to greater efficiency, specifically towards case priorities, with the ultimate benefit of cost savings and overall better case management.

1.3. Problem or Gap Identification

Law enforcement currently lacks real-time insights that indicate the probability of resolving a reported crime case. This absence of predictive intelligence not only affects investigative efficiency but also has a profound impact on victims. At the same time, many victims, particularly those unfamiliar with legal procedures, struggle with uncertainty—unsure whether filing a case is worthwhile, especially when they lack suspect information or substantial evidence. Without a system in place to prioritize cases based on their likelihood of resolution, victims are often left in the dark about their case status, receiving little to no updates from authorities. Cases lacking strong leads are often deprioritized, leaving victims uncertain about whether their pursuit of justice is worthwhile (Jellema, 2021). This lack of communication creates feelings of neglect, discouraging victims from pursuing justice, and weakening public trust in the investigative process.

To bridge this gap, we propose a predictive analytics model that classifies cases based on resolution likelihood, aiding law enforcement in strategic decision-making while equipping victims with realistic expectations. By leveraging machine learning, authorities can assess case solvability in real-time, enabling better prioritization and communication. Transparency ensures victims remain informed throughout investigations, reducing frustration and strengthening confidence in the justice system. This proactive approach enhances case management, fosters informed decision-making, and improves trust between law enforcement and the community.

1.4. Proposed Analytical Solution

The solving of criminal cases is determined by a broad array of interdependent variables. Knowledge of these variables is important to the enhancement of investigative procedures and the improvement of the efficacy of crime solving. To address this challenge, we propose the development of a binary classification model designed to predict whether a crime will be solved or unresolved.

This model uses significant attributes, including offenseType, SuspectKilled, SuspectOccupation, typeofPlace_Categorized, IncidentType, IncidentTypeSpecific_Categorized, VictimGender, VictimKilled, and VictimOccupation to identify patterns associated with case outcomes. By utilizing CaseStatus as the target variable, the model will provide data-driven classifications based on historical crime records and observed trends.

This data-driven approach will help victims understand the process better, improves communication between them and the police, and builds trust by providing clearer expectations about the case outcomes. By uncovering trends and patterns in crime data, our model has the potential to revolutionize decision-making and improve case solvability rates.

1.5. Expected Impact

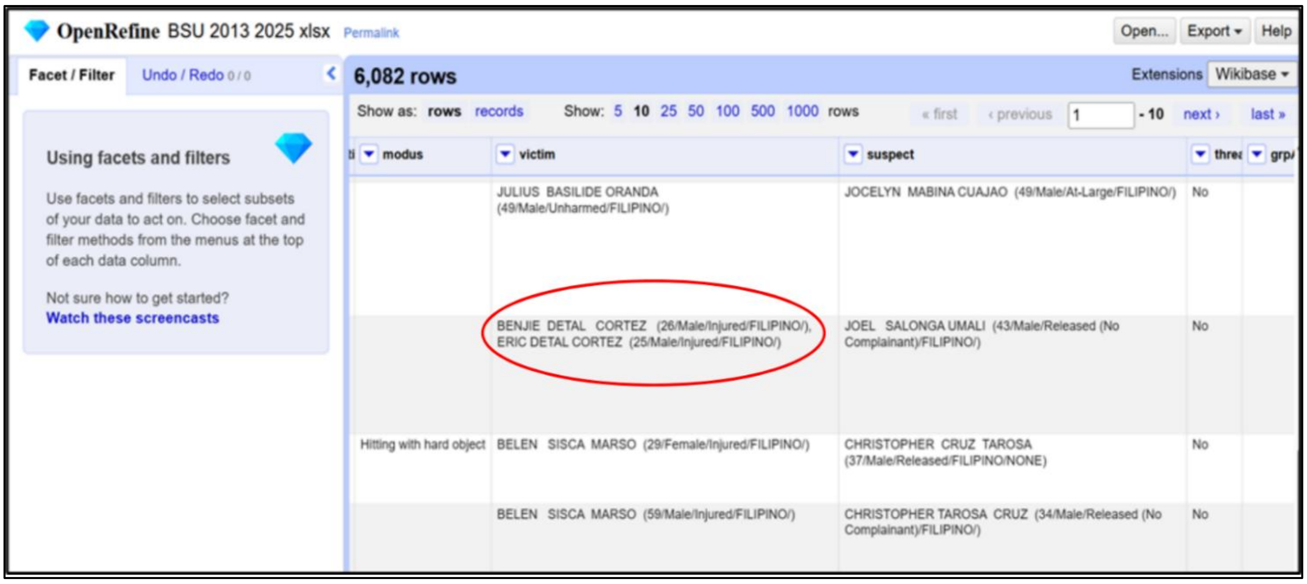
This predictive model serves as an essential tool for both victims and law enforcement, offering valuable insights into resolution likelihood. By analyzing investigative variables, the model allows police to allocate resources effectively—prioritizing solvable cases while adjusting strategies for more complex investigations. The model enhances decision-making, improves case solvability rates, and ultimately increases the chances of delivering justice to victims. Additionally, it empowers victims-especially those unfamiliar with legal procedures—by providing them with realistic expectations and guiding them through the justice system with greater confidence.

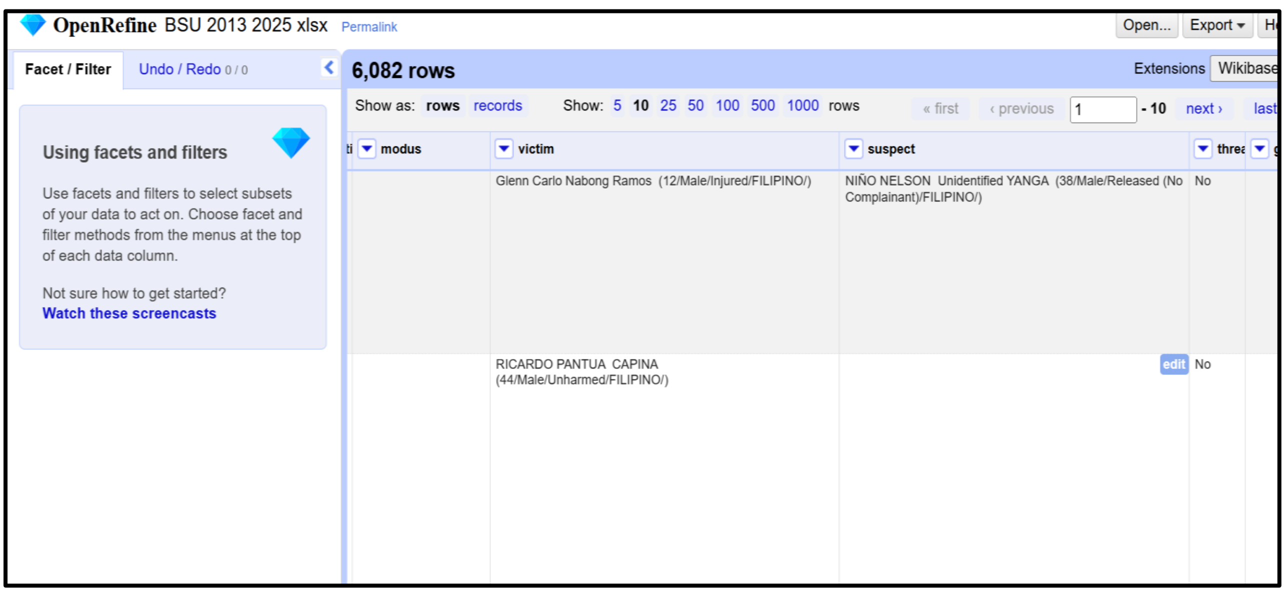
2. Preprocessing Techniques

OPEN REFINE

Figure 1

Split Multivalued Victims and Suspect

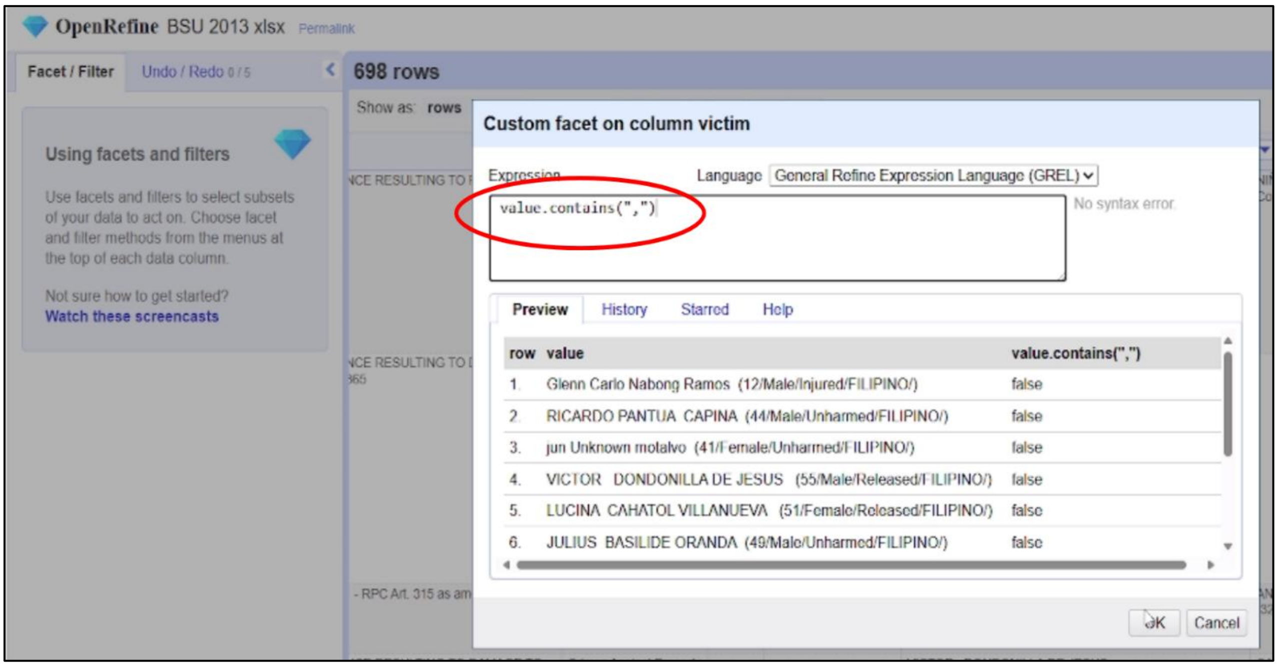




Splitting records that contain multiple instances in either the victim or suspect columns into individual rows for each entry.

Figure 2

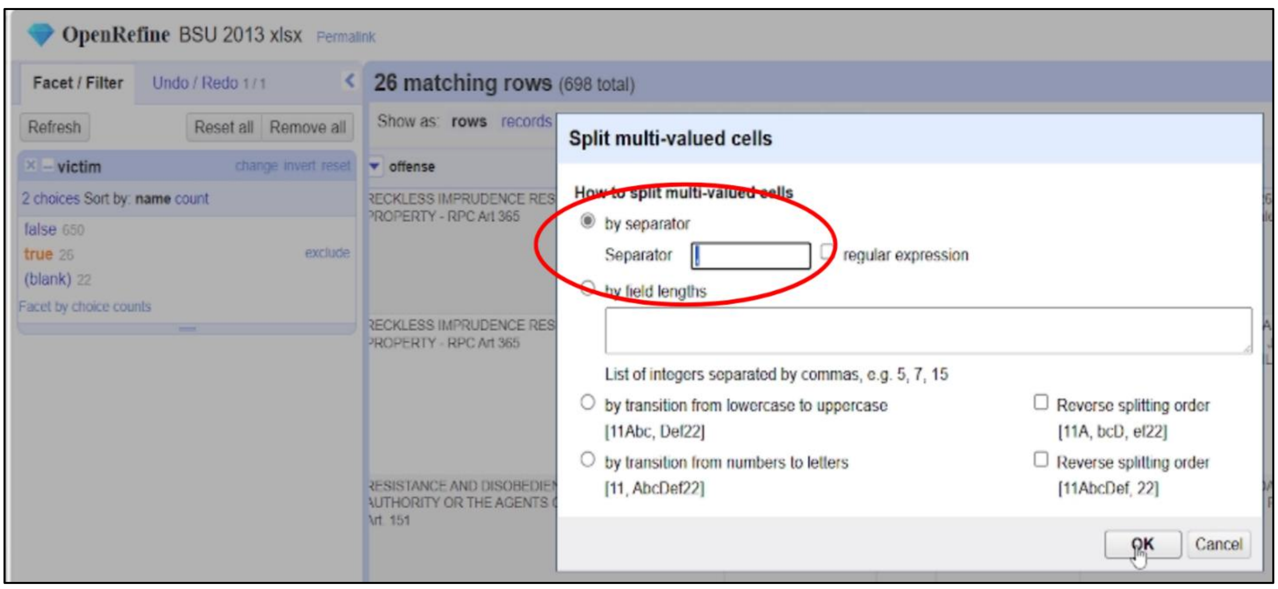
Custom Facet to Search for Multivalued Cells



This figure shows how to split the information of a victim and suspect using conditional formatting. The raw dataset used comma (",") to input multiple people in a single cell. By creating a custom facet, it will only display multivalued cells making it easy to check and split multivalued attributes.

Figure 3

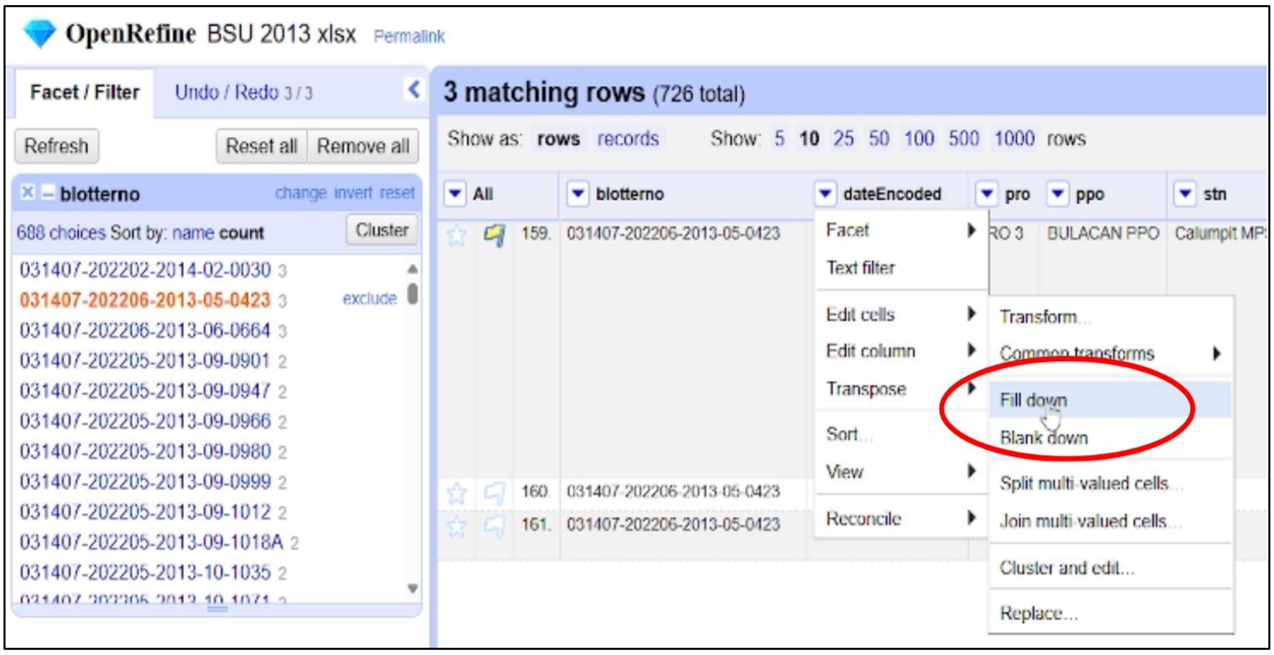
Split the Text Based on the Separator

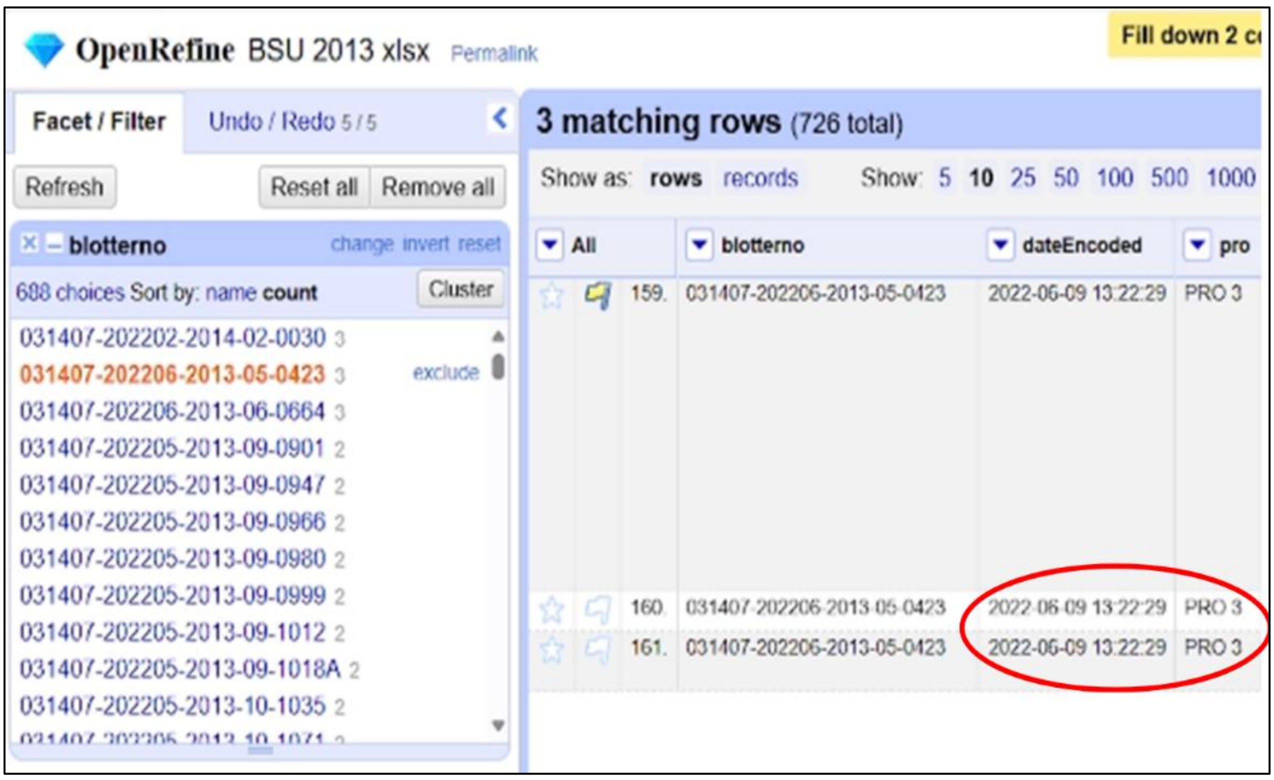


This figure shows how we split the text based on a separator. The operator used is a comma (",") for splitting the text into distinct parts.

Figure 4

Fill Down with Same Blotter No.



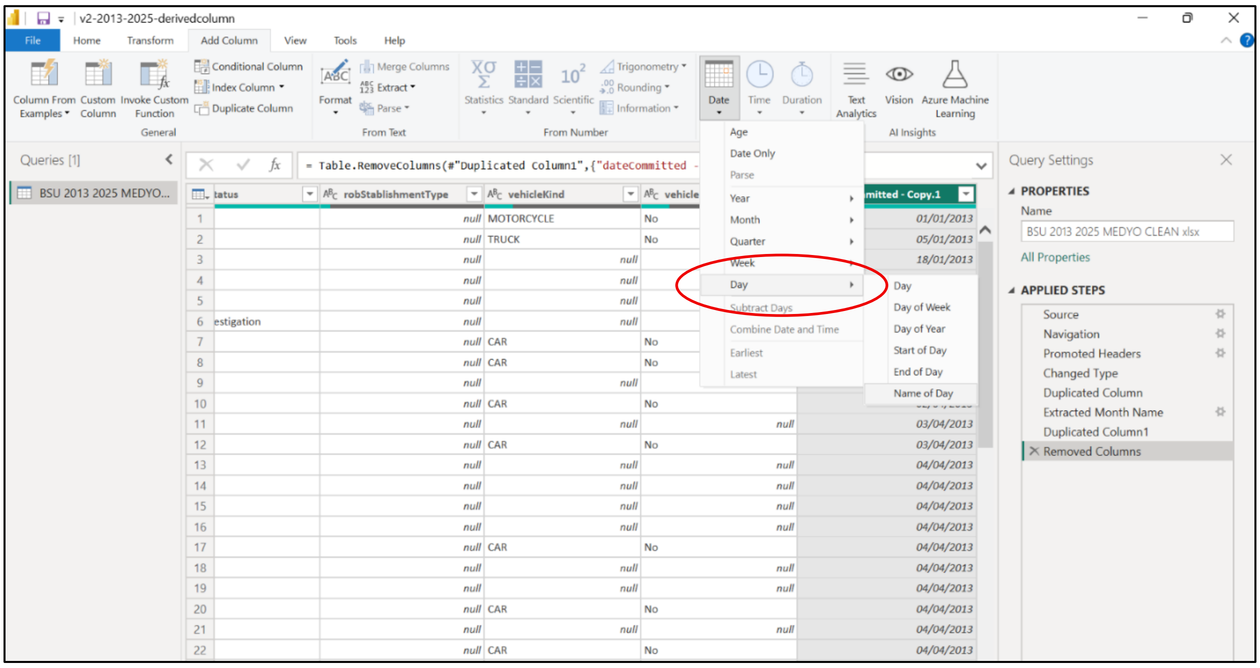


Using the "Fill down" method to copy values from the cell above into empty cells with the same Blotter No. The value in the first cell is automatically copied down to the cells below, making all rows under that column share the same Blotter No. without needing to manually enter it. This helps keep the data complete after editing or splitting rows.

POWER BI

Figure 5

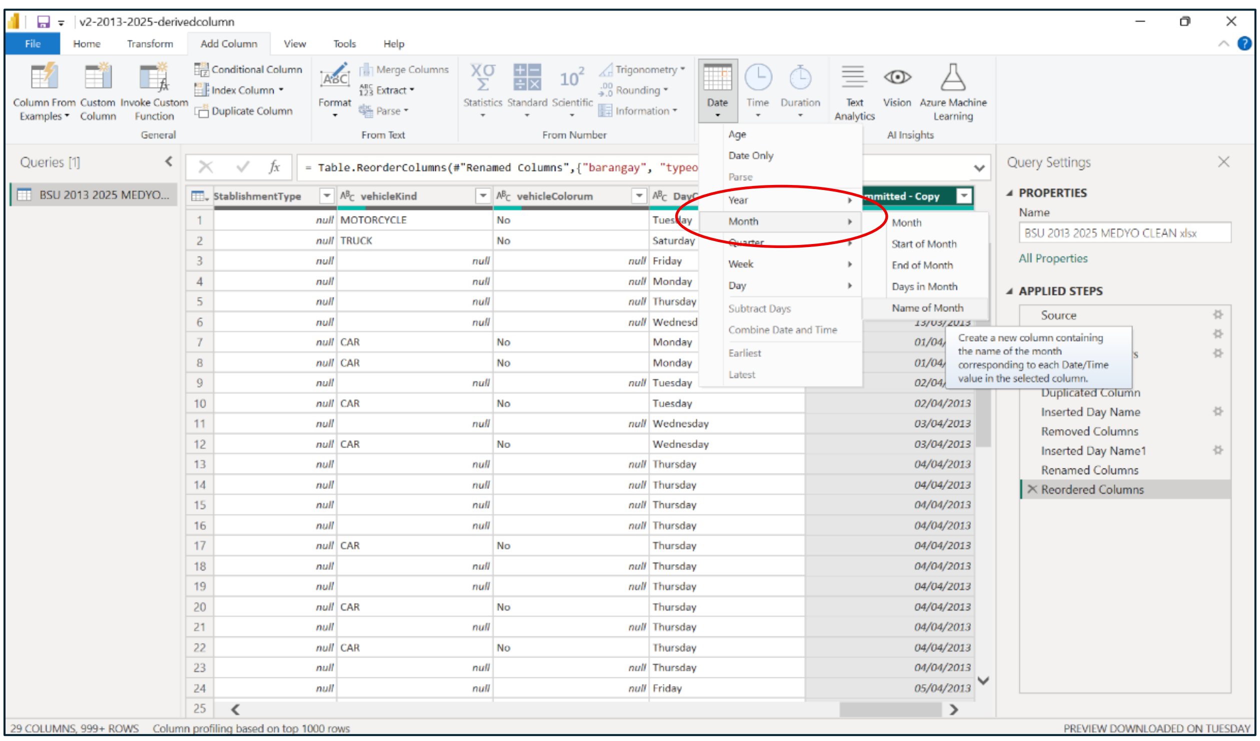
Derived Attribute (Day Committed)



The figure shows the derived attribute of *DayCommitted*, created from the attribute *DateCommitted*. This attribute will be helpful in analyzing trends and patterns in the daily occurrence of accidents in the Municipality of Calumpit.

Figure 6

Extracting MonthComitted from DateComitted

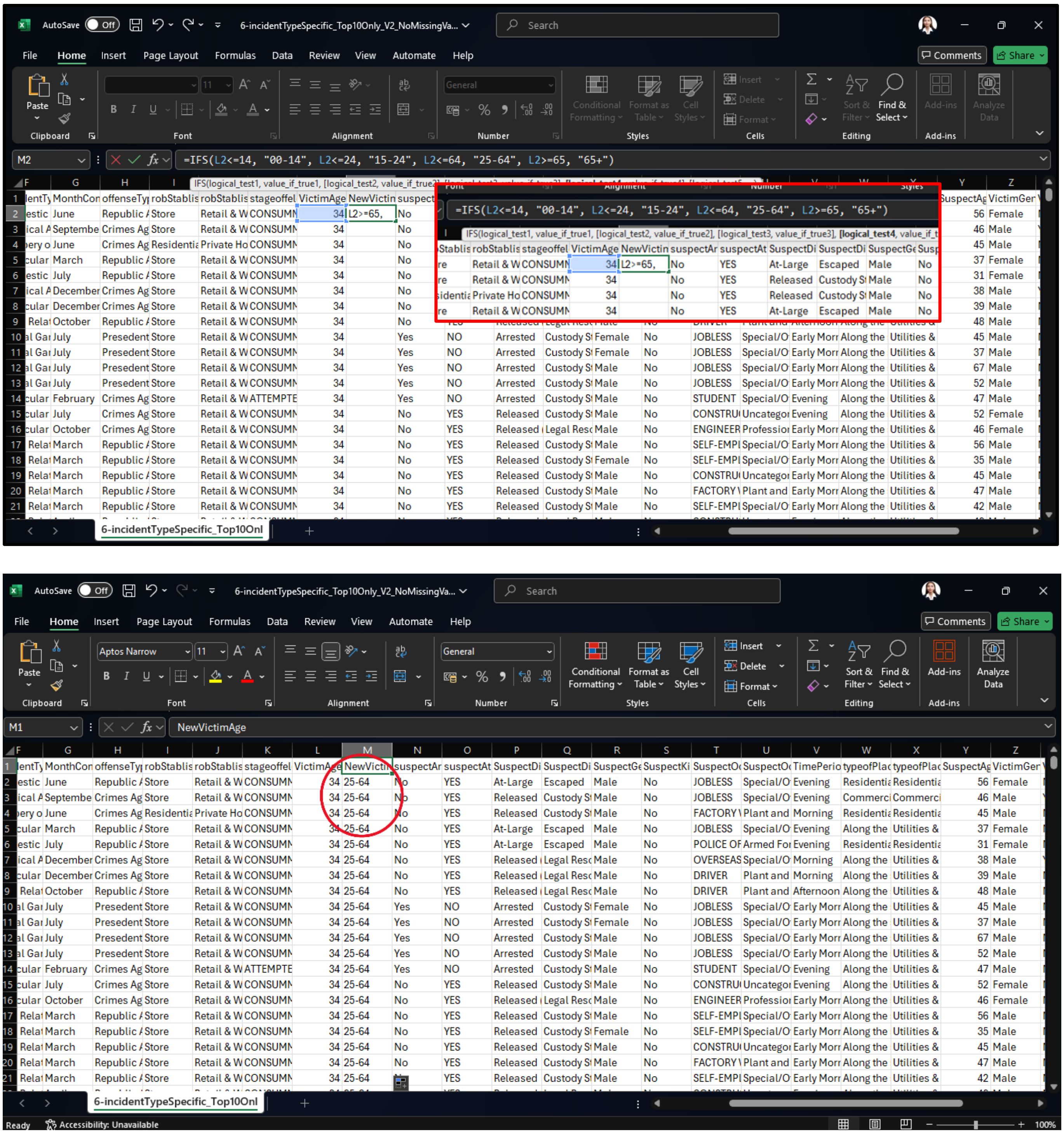


In Figure 6, the column *MonthCommitted* was extracted from the attribute *DateCommitted*. By utilizing this attribute, it will be possible to provide an insightful overview to determine which months have higher or lower rates of accidents in Calumpit.

Excel

Figure 7

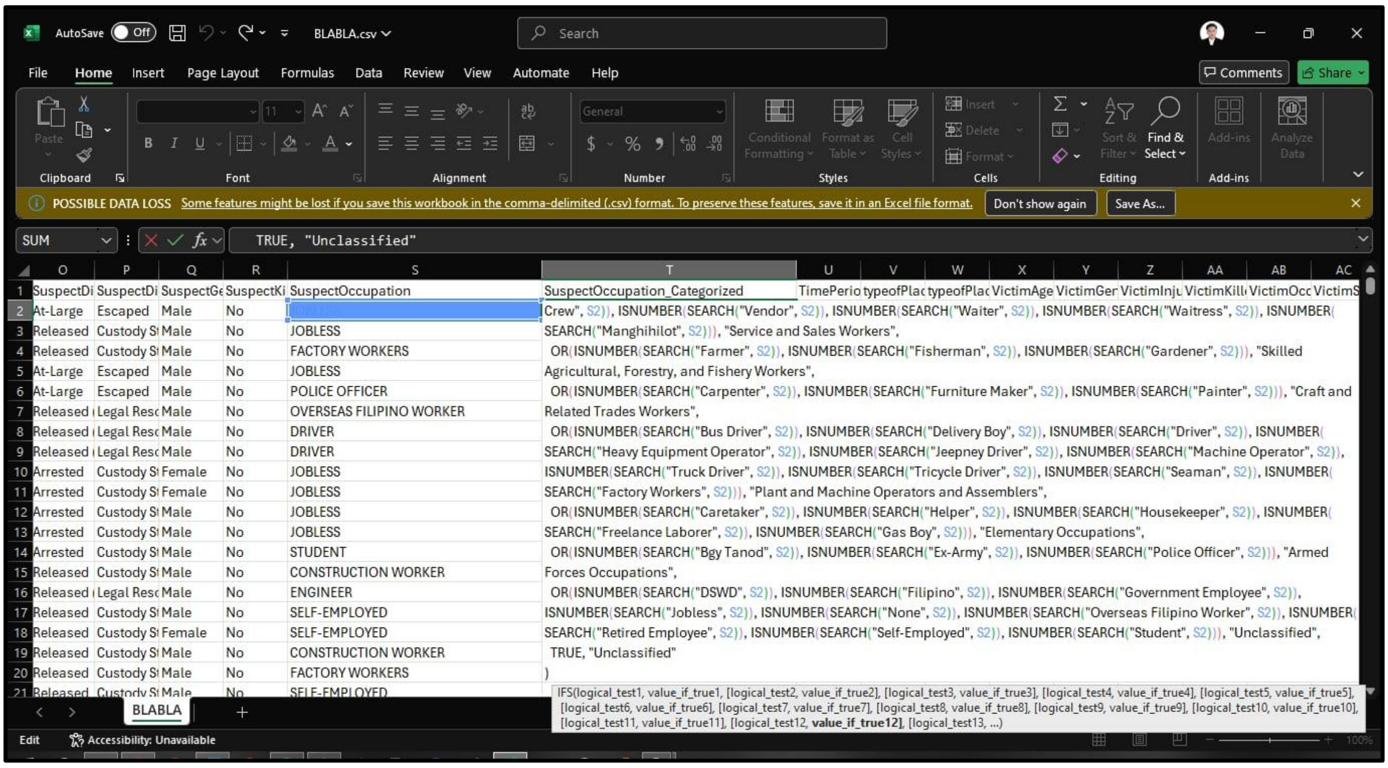
Binning or Categorizing Values

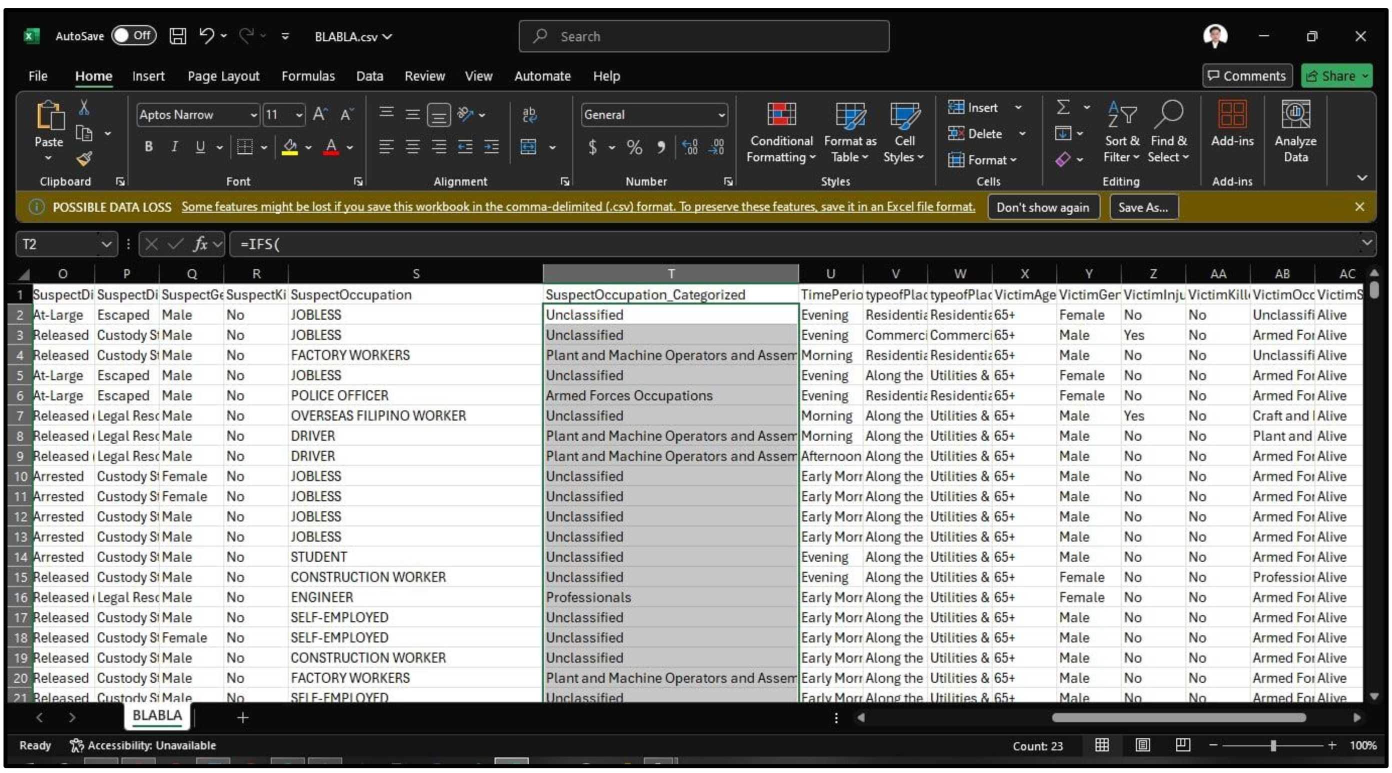


In addition to supporting analysis and interpretation of population trends, the age characteristic was discretized into various groups. The process involved transformation of age as a continuous characteristic into four distinct groups: 00–14, 15–24, 25–64, and 65+. With the use of Microsoft Excel, discretization was achieved using conditional formulas that considered the age of each individual and placed it in the corresponding group. With the help of Figure 7.1, this allowed for systematic grouping of ages, thus easier observations during the analysis of crime trends throughout the life stages. Discretization of age in this manner also helped reduce dataset variability and made visualization and modeling more effective for the predictive process. This discretization is done for both VictimAge and SuspectAge column for uniformity.

Figure 8

Categorizing SuspectOccupation with Specific Condition

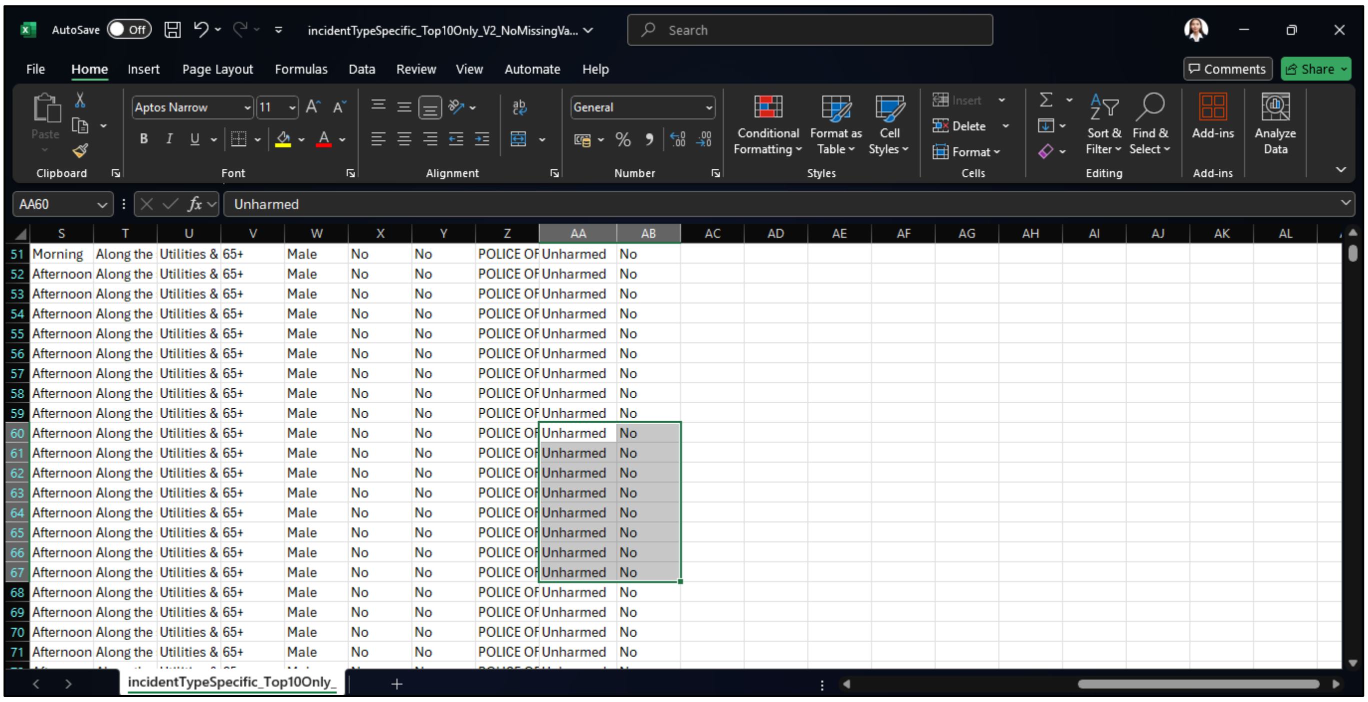


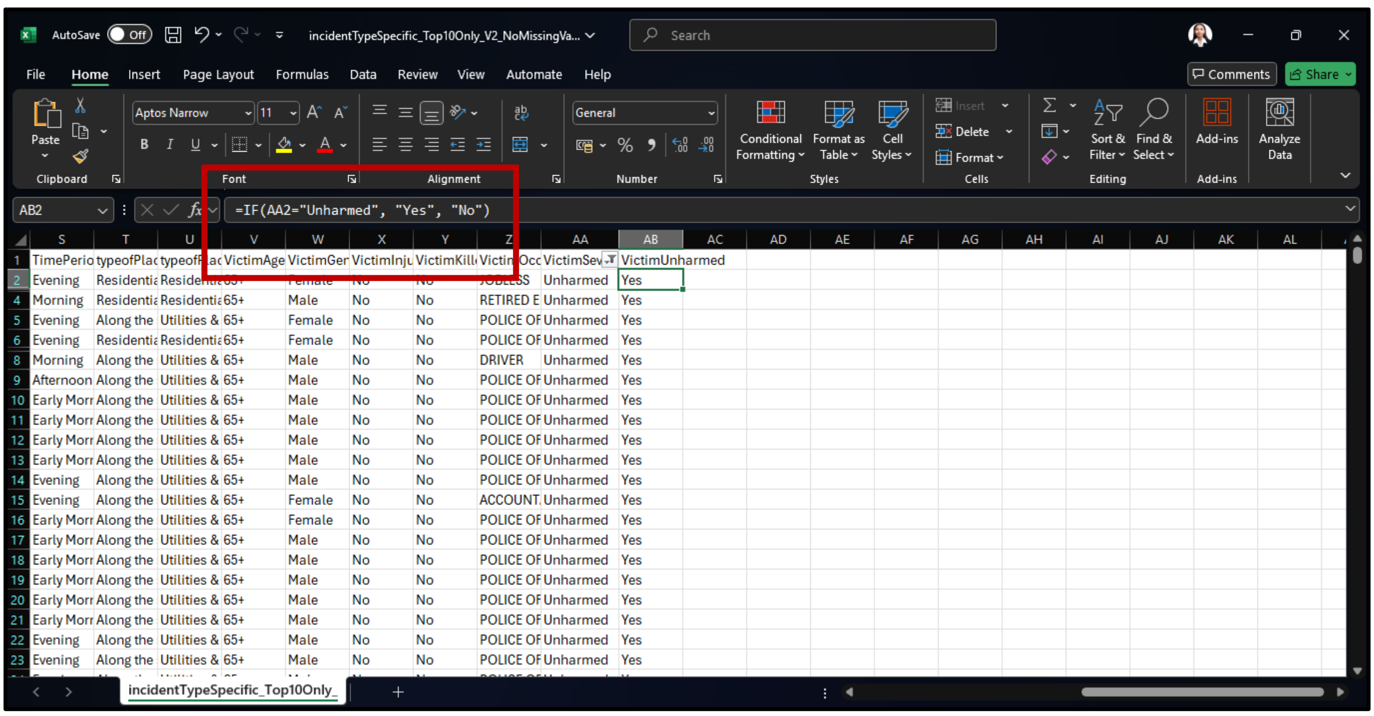


In this figure, the Excel sheet shows a formula being used to classify suspect occupations into broader categories. The SuspectOccupation_Categorized column uses an IFS() function with SEARCH() and ISNUMBER() to match keywords in the SuspectOccupation field (e.g., “Driver”, “Jobless”, “Police Officer”) and assign them to generalized groups like “Unclassified”, “Armed Forces Occupations”, or “Plant and Machine Operators”. In the same way, the following variables are also categorized for uniformity and to control unique values: IncidentTypeSpecific, typeofPlace, robStablishmentType, and SuspectDisposition.

Figure 9

Filtering Irrelevant or Noisy Records & Standardizing VictimUnharmed

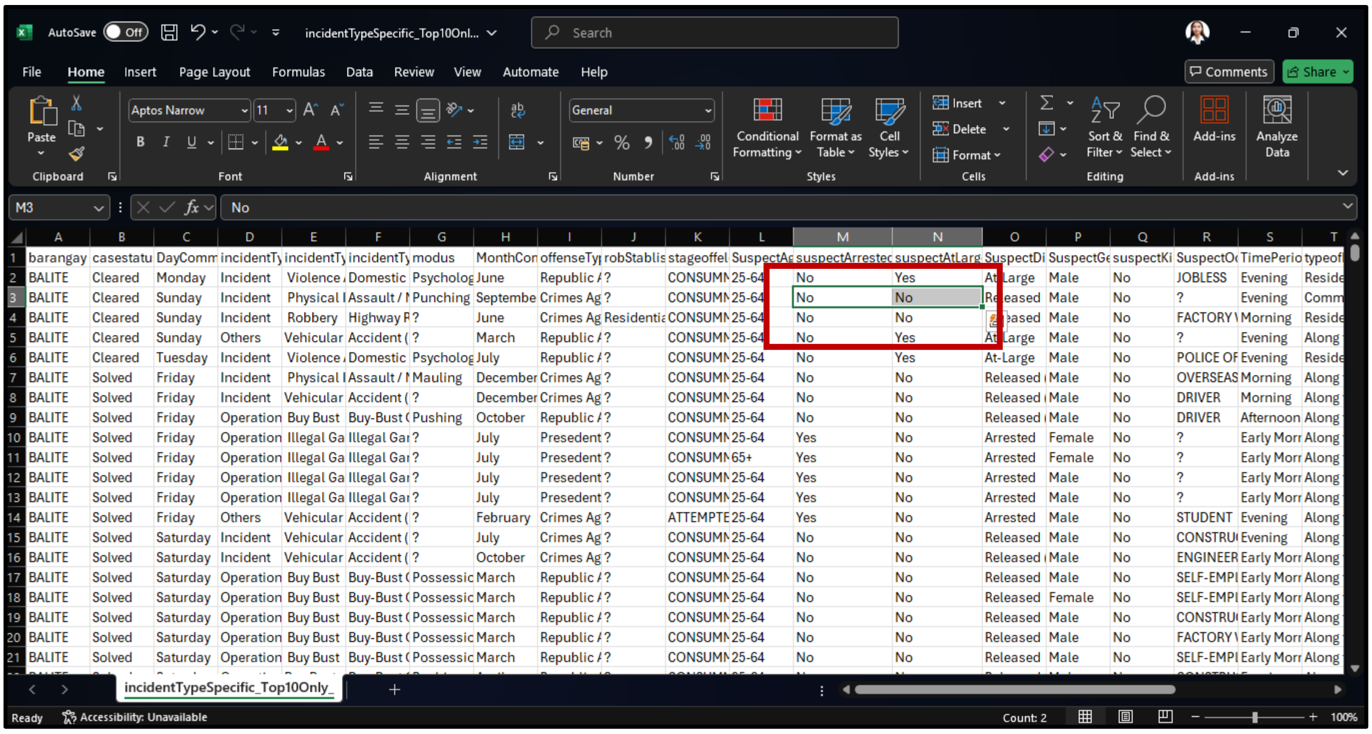


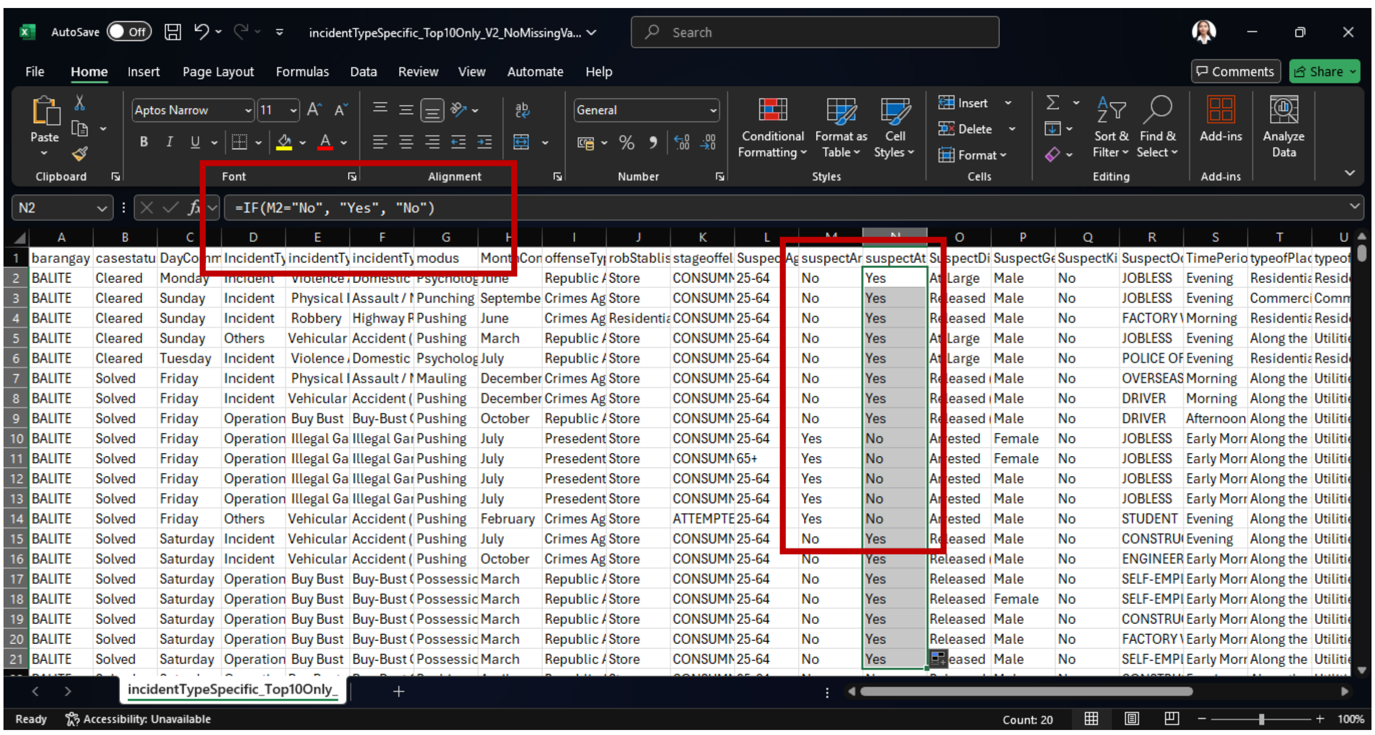


In this figure, the data cleaning process standardizes the `VictimUnharmed` field by converting every record where `VictimSeverity` equals “Unharmed” into “Yes” and all other records into “No.” This transformation aligns the two related fields, removes ambiguity, and ensures that the dataset uses a clear, binary format for analysis. By applying this rule across all columns, inconsistencies are eliminated, and the data becomes more reliable for downstream reporting or modeling.

Figure 10

Standardizing SuspectAtLarge



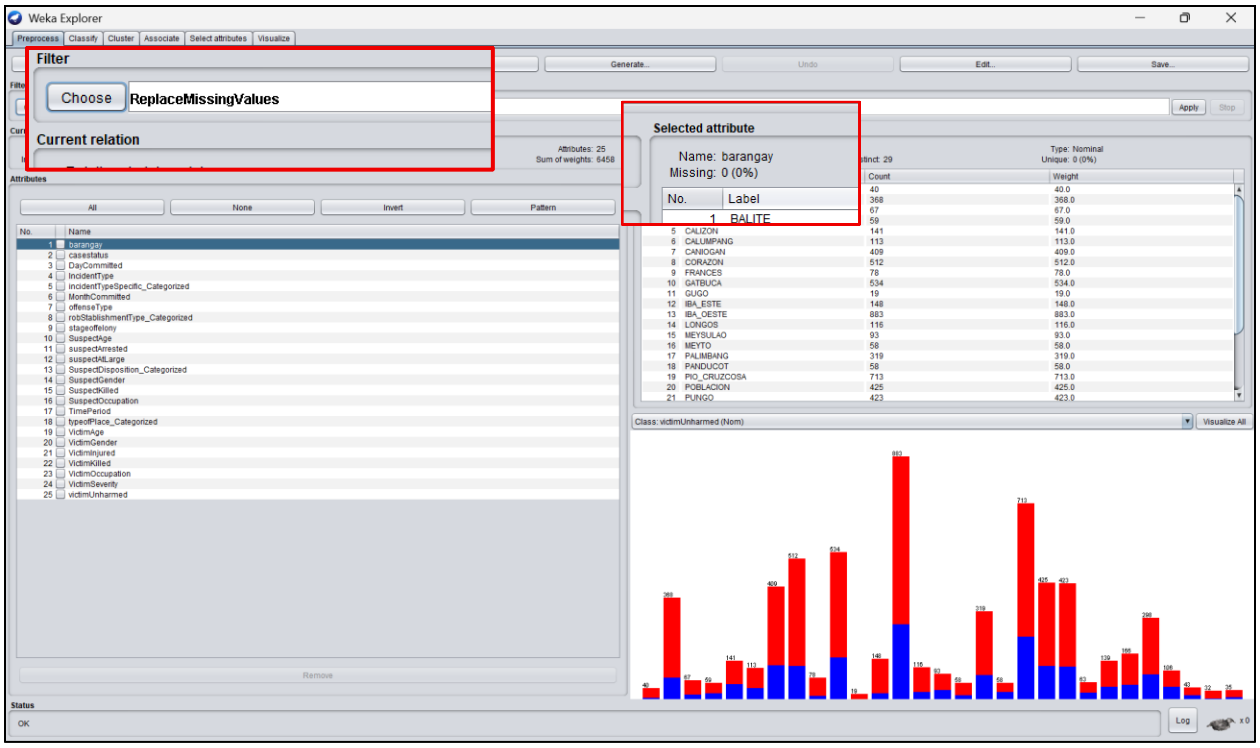


In the provided figure, the data cleaning process provides consistency between the columns SuspectArrested and SuspectAtLarge. The formula applied checks whether SuspectArrested equals "No," and if so, it puts "Yes" into SuspectAtLarge, signifying that the suspect is at large. If SuspectArrested is not "No" (that is, the suspect has been arrested), it puts "No" into SuspectAtLarge. This correction makes the connection between the two columns official, thus ensuring that the status of a suspect being at large is correctly represented based on their arrest status. The outcome gives cleaner and more dependable data for additional analysis.

WEKA

Figure 11

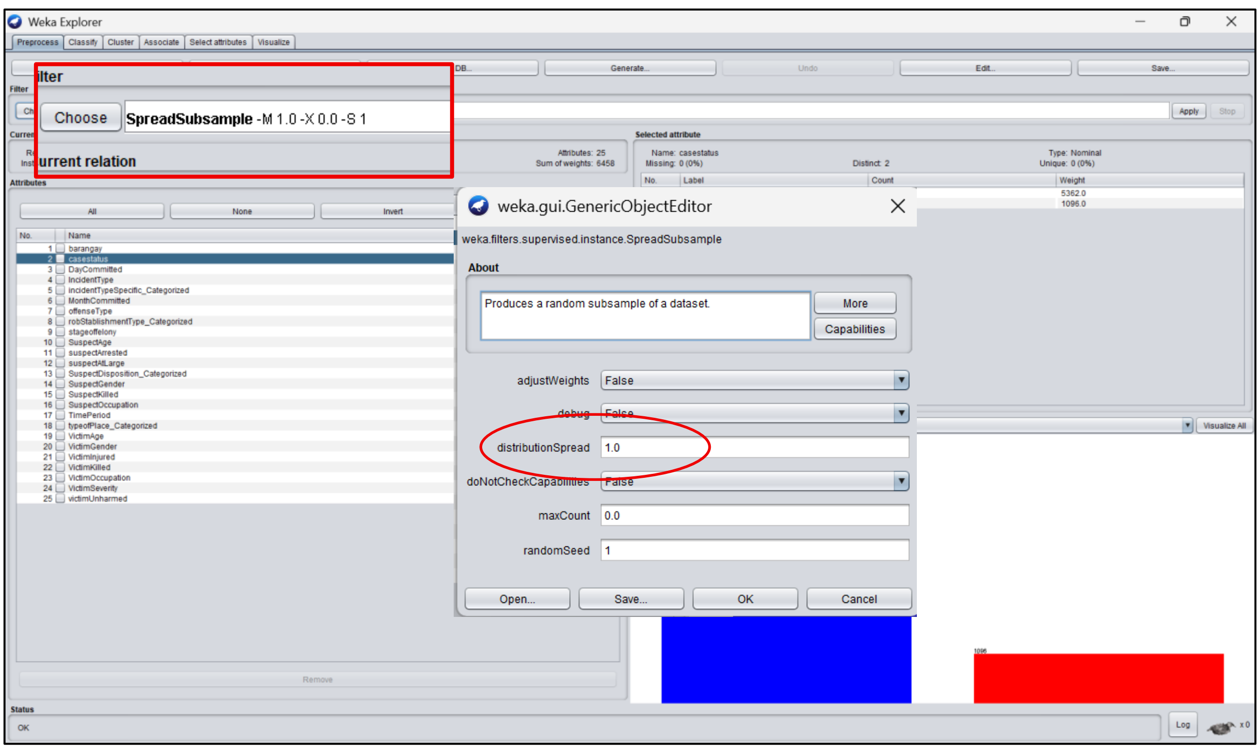
Imputation of Missing Values



The figure shows the imputation of values using WEKA's ReplaceMissingValues filter in the preprocess tab. The filter replaces the dataset's (all columns but skips the target class) nominal and numerical values with the modes and means of the columns.

Figure 12

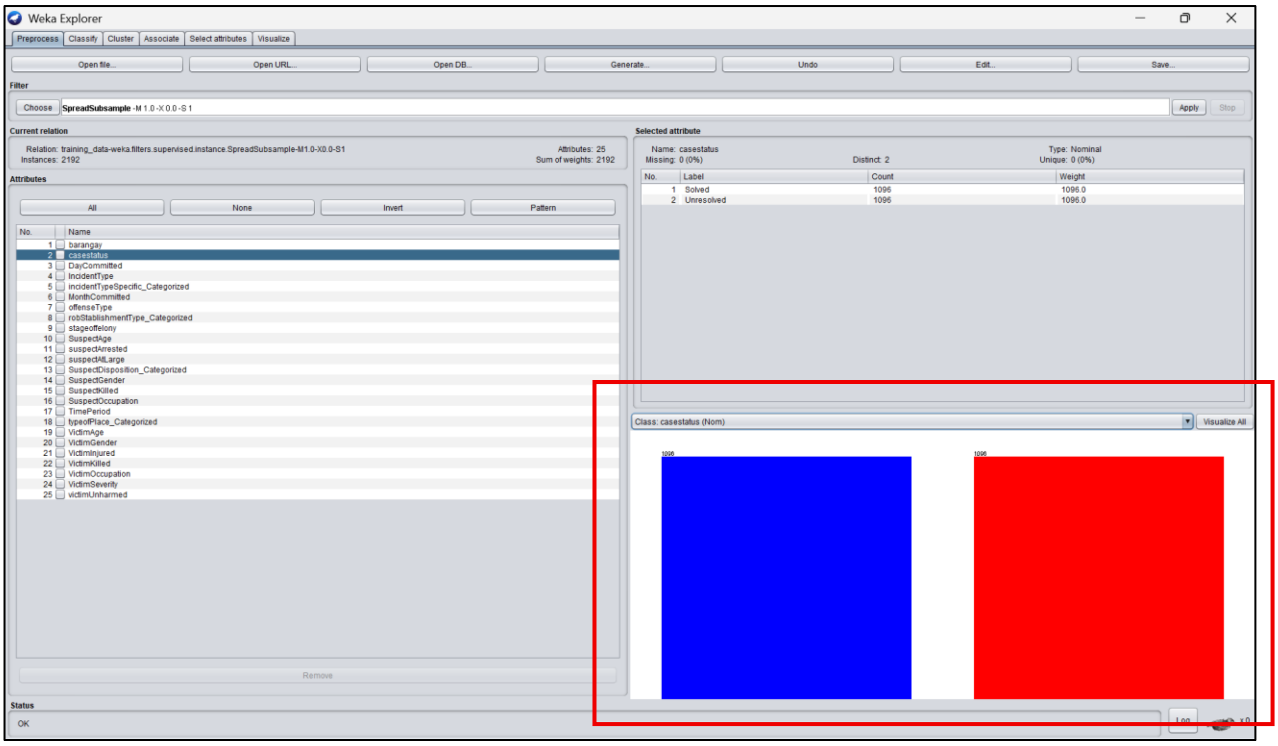
SpreadSubSample the Target Variable



The figure shows that we used RandomSpreadSubsample and set the distribution spread to 1.0 to balance the dataset. This parameter controls the class distribution after resampling. By setting it to 1.0, we ensured an even distribution among classes.

Figure 13

Balanced Target Class



This figure shows that after applying RandomSpreadSubsample, the class distribution for 'case status' was balanced, with each class having 1,096 instances.

3. Attribute Selection

Data Dictionary

Table 1

Attribute Names	**Description**	**Original Data Type / Measure**	**Original Values**	**Preprocessed Values**	**Data Type of the processed values**
1. blotterno	Unique blotter number assigned to a report, serving as a unique identifier for tracking cases.	Nominal	031407-202201-2014-01-0001	Not Applicable	Nominal
2. dateEncoded	Date when the report was entered into a system, indicating when	Numeric	2022-01-12	Not Applicable	Numeric

	the incident was officially recorded.				
3. pro	Police Regional Office responsible for overseeing law enforcement activities within a specific region.	Nominal	PRO 3	Not Applicable	Numeric
4. ppo	Provincial Police Office managing police operations and cases at the provincial level.	Nominal	BULACAN PPO	Not Applicable	Nominal
5. stn	Station where the report was recorded, representing the local police facility handling the case.	Nominal	Calumpit MPS	Not Applicable	Nominal
6. pcp	Municipal Police Station, referring to the local police station responsible for law enforcement within a municipality.	Nominal	MPS	Not Applicable	Nominal
7. region	Geographic region where the incident occurred, categorizing the location based on administrative divisions.	Nominal	CENTRAL LUZON	Not Applicable	Nominal
8. province	Province where the incident happened, specifying the larger territorial unit of occurrence.	Nominal	BULACAN	Not Applicable	Nominal
9. municipal	Municipality where the event took place, identifying the local government area involved.	Nominal	CALUMPIT	Not Applicable	Nominal
10. barangay	Specific barangay location of the event, pinpointing the smallest administrative unit where the crime occurred.	Nominal	IBA O'ESTE	Not Applicable	Nominal
11. typeofPlace	Type of place where the incident occurred (e.g., residential, commercial), describing the nature of the crime scene.	Nominal	Residential (house/condo)	Residential	Nominal

12. dateReported	Date when the incident was reported, marking when authorities were first informed.	Numeric	2013-01-01	Not Applicable	Numeric
13. timeReported	Time when the incident was reported, specifying the time and minute the report was filed.	Numeric	01:36 AM	Not Applicable	Numeric
14. dateCommitted	Date when the crime happened, indicating when the offense took place.	Numeric	2013-01-01	Not Applicable	Numeric
15. timeCommitted	Time of the crime, documenting the specific hour and minute of the event.	Numeric	12:36 AM	Early Morning, Morning, Afternoon, Evening	Nominal
16. incidentType	Type or classification of the incident, categorizing the nature of the offense.	Nominal	Incident, Others, Operations, Simple	Incident, Others, Operations	Nominal
17. iscime	Boolean flag for whether an incident is a crime, determining if it is legally classified as an offense.	Boolean	YES, NO	Not Applicable	Boolean
18. stageoffelony	Stage of the felony (attempted, consummated, frustrated), describing the progress of the crime.	Nominal	CONSUMMATED	Not Applicable	Nominal
19. offense	Specific offense committed, detailing the legal violation involved.	Nominal	LESS SERIOUS PHYSICAL INJURIES - RPC Art. 265	Not Applicable	Nominal
20. offenseType	Classification of the offense, categorizing it based on legal definitions and severity.	Nominal	Crimes Against Persons	Crimes Against Persons	Nominal
21. Modus	Modus operandi or method used in the crime, describing how the offense was carried out.	Nominal	Hitting with hard object	Not Applicable	Nominal
22. victim	Information about the victim(s), including personal details and their involvement in the case.	Nominal	Juan Dela Cruz (12/Male/Injured/FILIPINO/)	VictimAge, VictimGender, VictimSeverity, VictimOccupation, VictimNationality	Nominal

23. suspect	Information about the suspect(s), providing details on the individuals accused of the crime.	Nominal	Pedro Dela Cruz (38/Male/Released (No Complainant)/FILIPINO/)	SupectAge, SupectGender, SuspectDisposition, SuspectNationality, SuspectOccupation	Nominal
24. threatGrp	Threat group involved, if applicable, identifying any organized group linked to the incident.	Boolean	YES, NO	Not Applicable	Boolean
25. suspectisEGO	Whether the suspect belongs to an identified entity/government organization, marking official affiliations.	Boolean	YES, NO	Not Applicable	Boolean
26. victimisEGO	Whether the victim belongs to an identified entity/government organization, confirming official connections.	Boolean	YES, NO	Not Applicable	Boolean
27. victimKilled	Whether a victim was killed, indicating if the crime resulted in a fatality.	Boolean	YES, NO	Not Applicable	Boolean
28. victimInjured	Whether a victim was injured, specifying if any harm was inflicted.	Boolean	YES, NO	Not Applicable	Boolean
29. victimUnharmed	Whether a victim remained unharmed, confirming if the victim escaped without injury.	Boolean	YES, NO	Not Applicable	Boolean
30. suspectKilled	Whether the suspect was killed, indicating if law enforcement or other circumstances led to their death.	Boolean	YES, NO	Not Applicable	Boolean
31. suspectAtLarge	Whether the suspect is still at large, marking if they remain unapprehend.	Boolean	YES, NO	Not Applicable	Boolean
32. suspectArrested	Whether the suspect was arrested, confirming if law enforcement has taken them into custody.	Boolean	YES, NO	Not Applicable	Boolean
33. narrative	A detailed report of the incident, providing a comprehensive narrative of events.	Nominal	INITIAL INVESTIGATION REVEALED THAT THE TOYOTA INNOVA WAGON PARKED NEAR TO THE BASKETBALL	Not Applicable	Nominal

			COURT ACCIDENTALLY HIT BY THE BALL THE WINDSHIELD		
34. casestatus	Current status of the case, indicating whether it is open, closed, or under investigation.	Nominal	Cleared Solved Under Investigation	Solved Unresolved	Nominal
35. caseSolveType	How the case was resolved, specifying if it was solved, dismissed, or remains active.	Nominal	SOLVED (AMICABLY SETTLED)	Not Applicable	Nominal
36. caseStatusinProsec	Status of the case in prosecution, detailing if it has proceeded to legal action.	Nominal	FOR FURTHER INVESTIGATION	Not Applicable	Nominal
37. lat	Latitude coordinate of the incident, providing geolocation data for mapping purposes.	Numeric	14.897424	Not Applicable	Numeric
38. lng	Longitude coordinate of the incident, complementing latitude to pinpoint the exact location.	Numeric	120.765694	Not Applicable	Numeric
39. victimCount	Number of victims, indicating how many people were affected.	Numeric	1, 2, 3	Not Applicable	Nominal
40. suspectCount	Number of suspects, specifying the total individuals involved as perpetrators.	Numeric	1, 2, 3	Not Applicable	Nominal
41. investigator	Investigator assigned to the case, naming the officer responsible for gathering evidence.	Nominal	PSSG EDUARD A TENA - 09187771645	Not Applicable	Nominal
42. headInves	Head investigator, identifying the lead officer overseeing the case.	Nominal	PCMS RICKY S FANGON	Not Applicable	Nominal
43. placeCommission	Place where the crime was committed, specifying the exact site of the offense.	Nominal	Frances Elementary School	Not Applicable	Nominal
44. robStablishmentType	Type of establishment robbed (e.g., bank, store), describing the	Nominal	College/School/ Universty	EDUCATION	Nominal

	nature of the targeted property.				
45. vehicleKind	Type of vehicle (car, motorcycle, truck), classifying the mode of transport involved.	Nominal	Motor	Not Applicable	Nominal
46. vehicleColorum	Whether the vehicle is an unregistered public transport vehicle, identifying if it was used unlawfully for transport services.	Nominal	YES, NO	Not Applicable	Boolean

Note: The following attributes were removed from the dataset due to the absence of recorded values. Since they did not contain any meaningful information, they were excluded to ensure data integrity and relevance in analysis. By eliminating these attributes, the dataset remains streamlined, preventing unnecessary clutter while maintaining accuracy and efficiency in handling and processing the information.

Excluded Attributes: Street, grpAffiliation, incidenttypethreatgrp, mrs, suspectEGOPosition, suspectEGOClass, victimEGOPosition, victimEGOclass, suspectMotive, suspectMotive, SuspectSubMotive, groundDismissedProsec, fillingType, dateFiledtoProsec, ISDocketNo, prosecutor, caseCourtStatus, dateFiledInCourt, ccNo, judge, court, groundDismissedCourt, robEstablishmentName, faserialno, facaliber, fakind, famake, fastatus, drugInvolved, vehiclePlateNo, vehicleMCPlateNo, vehicleChassisNo, vehicleMake, vehicleModel, vehicleStatus, vehicleRegistrationNo, vehicleRegistrationStatus.

Evaluator + Search Methods

Table 2

Evaluator	Search Method	Significant Attributes (Arrange based on significance)
1. CfsSubsetEval	BestFirst (Full Training set)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 18 – typeofPlace_Categorized
2. CfsSubsetEval	BestFirst (Cross-validation)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 18 – typeofPlace_Categorized
3. CfsSubsetEval	GreedyStepWise (Full Training set)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 18 – typeofPlace_Categorized
4. CfsSubsetEval	GreedyStepWise (Cross-validation)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 18 – typeofPlace_Categorized
5. ClassifierAttributeEval	Ranker (Full Training set)	25 – victimUnharmed 24 – VictimSeverity

		9 – Stageoffelony 10 – SuspectAge 11 – suspectArrested 8 – robStablishmentType_Categorized 7 – offenseType 6 – MonthCommitted 3 – DayComitted 4 – IncidentType 5 – IncidentTypeSpecific_Categorized 12 – suspectAtLarge 13 – SuspectDisposition_Categorized 14 – SuspectGender 22 – VictimKilled 23 – VictimOccupation 20 – VictimGender 21 – VictimInjured 19 – VictimAge 15 – SuspectKilled 16 – SuspectOccupation 17 – TimePeriod 18 – typeofPlace_Categorized 1 – barangay
6. ClassifierAttributeEval	Ranker (Cross-validation)	25 – victimUnharmed 24 – VictimSeverity 9 – Stageoffelony 10 – SuspectAge 11 – suspectArrested 8 – robStablishmentType_Categorized 7 – offenseType 6 – MonthCommitted 3 – DayComitted 4 – IncidentType 5 – IncidentTypeSpecific_Categorized 12 – suspectAtLarge 13 – SuspectDisposition_Categorized 14 – SuspectGender 22 – VictimKilled 23 – VictimOccupation 20 – VictimGender 21 – VictimInjured 19 – VictimAge 15 – SuspectKilled 16 – SuspectOccupation 17 – TimePeriod 18 – typeofPlace_Categorized 1 – barangay
7. ClassifierSubsetEval	BestFirst (Full Training set)	----
8. ClassifierSubsetEval	BestFirst (Cross-validation)	----
9. ClassifierSubsetEval	GreedyStepWise (Full Training set)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 12 – suspectAtLarge

		11 – suspectArrested 18 – typeofPlace_Categorized 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 20 – VictimGender 22 – VictimKilled 24 – VictimSeverity 8 – robStablishmentType_Categorized 15 – SuspectKilled 7 – offenseType 21 – VictimInjured 9 – Stageoffelony 14 – SuspectGender 23 – VictimOccupation 10 – SuspectAge 19 – VictimAge 25 – victimUnharmed 6 – MonthCommitted 17 – TimePeriod 1 – barangay 3 – DayComitted
10. ClassifierSubsetEval	GreedyStepWise (Cross-validation)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 12 – suspectAtLarge 11 – suspectArrested 18 – typeofPlace_Categorized 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 20 – VictimGender 22 – VictimKilled 24 – VictimSeverity 8 – robStablishmentType_Categorized 15 – SuspectKilled 7 – offenseType 21 – VictimInjured 9 – Stageoffelony 14 – SuspectGender 23 – VictimOccupation 10 – SuspectAge 19 – VictimAge 25 – victimUnharmed 6 – MonthCommitted 17 – TimePeriod 1 – barangay 3 – DayComitted
11. CorrelationAttributeEval	Ranker (Full Training set)	13 – SuspectDisposition_Categorized 12 – suspectAtLarge 11 – suspectArrested 15 – SuspectKilled 16 – SuspectOccupation 5 – IncidentTypeSpecific_Categorized 4 – IncidentType

		8 – robStablishmentType_Categorized 22 – VictimKilled 18 – typeofPlace_Categorized 24 – VictimSeverity 20 – VictimGender 7 – offenseType 9 – Stageoffelony 14 – SuspectGender 1 – barangay 23 – VictimOccupation 10 – SuspectAge 19 – VictimAge 6 – MonthCommitted 21 – VictimInjured 17 – TimePeriod 3 – DayComitted 25 – victimUnharmed
12. CorrelationAttributeEval	Ranker (Cross-validation)	13 – SuspectDisposition_Categorized 11 – suspectArrested 12 – suspectAtLarge 15 – SuspectKilled 16 – SuspectOccupation 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 8 – robStablishmentType_Categorized 22 – VictimKilled 18 – typeofPlace_Categorized 24 – VictimSeverity 20 – VictimGender 7 – offenseType 9 – Stageoffelony 14 – SuspectGender 1 – barangay 23 – VictimOccupation 10 – SuspectAge 19 – VictimAge 6 – MonthCommitted 21 – VictimInjured 17 – TimePeriod 3 – DayComitted 25 – victimUnharmed
13. GainRatioAttributeEval	Ranker (Full Training set)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 16 – SuspectOccupation 12 – suspectAtLarge 11 – suspectArrested 4 – IncidentType 18 – typeofPlace_Categorized 7 – offenseType 20 – VictimGender 1 – barangay 6 – MonthCommitted

		23 – VictimOccupation 22 – VictimKilled 15 – SuspectKilled 24 – VictimSeverity 8 – robStablishmentType_Categorized 19 – VictimAge 10 – SuspectAge 3 – DayComitted 17 – TimePeriod 21 – VictimInjured 9 – Stageoffelony 14 – SuspectGender 25 – victimUnharmed
14. GainRatioAttributeEval	Ranker (Cross-validation)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 16 – SuspectOccupation 11 – suspectArrested 12 – suspectAtLarge 4 – IncidentType 18 – typeofPlace_Categorized 7 – offenseType 20 – VictimGender 1 – barangay 6 – MonthCommitted 23 – VictimOccupation 22 – VictimKilled 15 – SuspectKilled 24 – VictimSeverity 8 – robStablishmentType_Categorized 19 – VictimAge 10 – SuspectAge 3 – DayComitted 21 – VictimInjured 17 – TimePeriod 9 – Stageoffelony 14 – SuspectGender 25 – victimUnharmed
15. InfoGainAttributeEval	Ranker (Full Training set)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 16 – SuspectOccupation 11 – suspectArrested 12 – suspectAtLarge 18 – typeofPlace_Categorized 4 – IncidentType 20 – VictimGender 7 – offenseType 23 – VictimOccupation 1 – barangay 6 – MonthCommitted 19 – VictimAge 10 – SuspectAge 21 – VictimInjured

		24 – VictimSeverity 25 – victimUnharmed 22 – VictimKilled 17 – TimePeriod 8 – robStablishmentType_Categorized 15 – SuspectKilled 14 – SuspectGender 9 – Stageoffelony 3 – DayComitted
16. InfoGainAttributeEval	Ranker (Cross-validation)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 16 – SuspectOccupation 11 – suspectArrested 12 – suspectAtLarge 18 – typeofPlace_Categorized 4 – IncidentType 20 – VictimGender 7 – offenseType 23 – VictimOccupation 1 – barangay 6 – MonthCommitted 19 – VictimAge 10 – SuspectAge 21 – VictimInjured 25 – victimUnharmed 17 – TimePeriod 24 – VictimSeverity 22 – VictimKilled 8 – robStablishmentType_Categorized 3 – DayComitted 15 – SuspectKilled 14 – SuspectGender 9 – Stageoffelony
17. OneRAttributeEval	Ranker (Full Training set)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 7 – offenseType 16 – SuspectOccupation 1 – barangay 23 – VictimOccupation 6 – MonthCommitted 11 – suspectArrested 12 – suspectAtLarge 3 – DayComitted 18 – typeofPlace_Categorized 17 – TimePeriod 19 – VictimAge 20 – VictimGender 22 – VictimKilled 15 – SuspectKilled 10 – SuspectAge 24 – VictimSeverity

		25 – victimUnharmed 21 – VictimInjured 8 – robStablishmentType_Categorized 14 – SuspectGender 9 – Stageoffelony
18. OneRAttributeEval	Ranker (Cross-validation)	13 – SuspectDisposition_Categorized 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 7 – offenseType 16 – SuspectOccupation 1 – barangay 23 – VictimOccupation 11 – suspectArrested 6 – MonthCommitted 12 – suspectAtLarge 18 – typeofPlace_Categorized 3 – DayComitted 17 – TimePeriod 19 – VictimAge 20 – VictimGender 22 – VictimKilled 15 – SuspectKilled 10 – SuspectAge 24 – VictimSeverity 21 – VictimInjured 25 – victimUnharmed 8 – robStablishmentType_Categorized 14 – SuspectGender 9 – Stageoffelony
19. PrincipalComponents	Ranker (Full Training set)	-----
20. ReliefFAttributeEval	Ranker (Cross-validation)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 12 – suspectAtLarge 11 – suspectArrested 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 18 – typeofPlace_Categorized 20 – VictimGender 7 – offenseType 22 – VictimKilled 15 – SuspectKilled 8 – robStablishmentType_Categorized 24 – VictimSeverity 1 – barangay 23 – VictimOccupation 6 – MonthCommitted 19 – VictimAge 10 – SuspectAge 9 – Stageoffelony 14 – SuspectGender 21 – VictimInjured 3 – DayComitted

		17 – TimePeriod 25 – victimUnharmed
21. ReliefFAttributeEval	Ranker (Full Training set)	13 – SuspectDisposition_Categorized 16 – SuspectOccupation 12 – suspectAtLarge 11 – suspectArrested 5 – IncidentTypeSpecific_Categorized 4 – IncidentType 18 – typeofPlace_Categorized 20 – VictimGender 7 – offenseType 22 – VictimKilled 15 – SuspectKilled 8 – robStablishmentType_Categorized 24 – VictimSeverity 1 – barangay 23 – VictimOccupation 6 – MonthCommitted 19 – VictimAge 10 – SuspectAge 9 – Stageoffelony 14 – SuspectGender 21 – VictimInjured 3 – DayComitted 17 – TimePeriod 25 – victimUnharmed
22. SymmetricalUncertAttributeEval	Ranker (Cross-validation)	----
23. SymmetricalUncertAttributeEval	Ranker (Full Training set)	----

Final list of selected attributes

Table 3

Attributes	Data Type / Measure
1. offenseType	Nominal
2. SuspectKilled	Boolean
3. SuspectOccupation	Nominal
4. typeofPlace_Categorized	Nominal
5. IncidentType	Nominal
6. incidentTypeSpecific_Categorized	Nominal
7. VictimGender	Nominal
8. VictimKilled	Boolean
9. VictimOccupation	Nominal
10. caseStatus	Nominal

In predictive analytics, selecting the right attributes is important to increase the model's accuracy, lowering overfitting, and enhancing interpretability. Table 3 highlights the selected attributes after conducting the attribute selection in Weka. The attributes were selected based on their applicability and contribution to the model's capability for prediction. This project's goal is to generate accurate predictions and streamline data processing by concentrating on these vital features.

An initial set of eleven (11) predictors or attributes was selected for model development. Upon further analysis using Weka's attribute evaluation method, it was observed that three attributes, although statistically strong, reduced the model's overall sensitivity (true positive rate). Namely, SuspectDisposition_Categorized, along with SuspectAtLarge and SuspectArrested, consistently ranked as top predictors across multiple evaluation techniques.

However, these attributes were removed from the final model to enhance sensibility and mitigate the risk of overfitting. Their direct relationship with case outcomes introduced potential data leakage, which could inflate performance metrics during training but reduce real-world applicability. Following their removal, IncidentType and SuspectKilled were introduced into the model. Both showed high predictive value and relevance based on attribute selection metrics, while maintaining the integrity and interpretability of the model. Offense type is a crucial attribute that is already part of the original attribute selection in Weka, and its inclusion is strongly supported by both analytical reasoning and research. In simple terms, offense type helps provide important context about a person's criminal behavior and plays a key role in predicting future actions—such as whether someone might commit another crime. Including offense type improves the accuracy of classification models by giving clear behavior or background. A study by Barnes et al. (2019) showed that offense type was one of the most important factors in predicting the risk of someone committing another violent act. The study highlighted that this kind of information makes machine learning models much more reliable for helping in legal decision-making. This proves that using offense type in Weka is both helpful and necessary for building better, smarter models in criminal justice.

In addition to understanding what kind of crime was committed, it is equally important to consider where it occurred. The attribute type of place categorized is also part of the original attribute selection in Weka and plays a vital role in analyzing the spatial aspects of criminal behavior. This approach is grounded in Crime Pattern (CP) theory and Rational Choice Theory, which provide insight into how offenders select crime locations. According to CP theory, individuals tend to commit crimes in areas where opportunities intersect with their awareness space —the locations they are familiar with through daily routines, such as home, school, work, or recreational sites (Brantingham & Brantingham, 1991, 1993). This connects to the concept of activity space, which includes places where people sped significant time and the routes between them (Brown & Moore, 1970; Horton & Reynolds, 1971; Golledge, 1978; Golledge & Stimson, 1997). By categorizing and analyzing the type of place, the model can better capture spatial patterns in crime, enhancing its ability to predict and understand criminal activity with greater precision.

Furthermore, IncidentTypeSpecific_Categorized—in simple terms, the crime type—was included as one of the predictors. Crime type is important because it shows that the kind of offense in a Domestic

Violence and Abuse environment has a direct and quantifiable effect on the results of criminal justice and enforcement (Barbin et al., 2025). This attribute is significant as it provides contextual insight into the nature of the crime, which can influence various outcomes such as case resolution, suspect behavior, and incident severity. Including this variable enhances the model's ability to identify patterns and improve classification accuracy across different crime categories.

Likewise, the gender of the victim is another attribute originally included in the selection and holds significant relevance in crime analysis. Previous research on homicide clearance has explored the influence of victim gender, though results have varied regarding its impact on whether a case is solved (Wolfgang, 1958; Riedel & Rinehart, 1996; Puckett & Lundman, 2003; Lee, 2005; Alderden & Lavery, 2007; Regoeczi et al., 2008). This variable is often linked to discretionary decision-making in the investigative process. To effectively represent this in machine learning models, victim gender is commonly encoded into two categories: female and male. Including this attribute allows for a more nuanced understanding of patterns in case outcomes and improves the model's ability to capture social and procedural dynamics within criminal investigations.

Moreover, the occupation of both offender and victim is considered as a strong predictor in this study because it can help to establish the pattern and relationship of the suspect and victim Reilly (2019). This variable may influence the behavior, the location of profession, and access to resources that is relevant to crimes. It assists with understanding the dynamics in the relationship, especially given that some of the victims know their offenders. The occupation, together with geographic and crime-related information, helps in behavior profiling and prediction, thus strengthening the investigative strategy. Lastly, if the victim was killed gives a statistically significant in association with specific type of killing. Thus, this attribute serves as a predictor because it directly reflects the severity and intent of the crime, making it crucial in distinguishing between types of violent offenses. These factors help investigator and researchers identify patterns across crimes.

4. **Selection Model**

Chosen study title: Analysis of Student Performance Using Data Mining Algorithm in Weka

This study aims to explore how data mining techniques can be applied to educational datasets to analyze and predict student performance. The goal is to identify patterns and factors that influence academic success, which can inform interventions to improve educational outcomes. In the study conducted by Zacharias and Manoj (2024), the top-performing classifiers identified were Random Tree, J48, and ZeroR. In alignment with their findings, the present study applied these three algorithms to the current dataset to evaluate their performance within a similar context.

1. Random Tree

(1) literature that used random tree: ***"Using the Random Tree Classifier to Improve the Project's Cost Predictability in Earned Value Management"***

The Random Tree algorithm is a decision tree method that builds a model by randomly selecting a subset of features at each split to reduce overfitting and improve generalization. In the study, it was used to analyze project data and predict cost performance more accurately than traditional methods, demonstrating its effectiveness in handling complex, structured data for reliable classification.

Nature of the study:

The study titled "Using the Random Tree Classifier to Improve the Project's Cost Predictability in Earned Value Management: An Empirical Study" focuses on enhancing the accuracy of cost forecasts in project management. Traditional Earned Value Management (EVM) metrics often fall short in predicting future cost deviations effectively. To address this, the researchers employed the Random Tree algorithm, a machine learning approach that introduces randomness in feature selection to reduce overfitting and improve prediction accuracy. The study analyzed real-world project data and demonstrated that the Random Tree model could outperform conventional methods by providing more stable and accurate predictions of whether a project would experience a cost overrun (Da S Fernandes & De Souza, 2019).

Relevance to our Study:

Although this study is centered on project cost prediction, its methodology is directly relevant to our research, which involves predicting the status of crime cases as either "Solved" or "Unresolved". Both studies deal with binary classification problems using structured data and aim to improve the accuracy of outcome predictions through data mining. The use of the Random Tree algorithm in their study supports our choice, as it proves effective in handling categorical and numerical data while reducing overfitting — a challenge we also face in crime data analysis. Therefore, their findings offer a strong methodological basis for our use of Random Tree in predicting case outcomes (Da S Fernandes & De Souza, 2019).

(2) literature that used random tree: **"Crime Prediction Using Data Mining and Machine Learning"**

The Random Tree algorithm is a machine learning method that builds a decision tree by selecting a random subset of features at each split, which helps reduce overfitting and improve prediction accuracy. In the study on crime prediction, it was used to analyze crime data and classify patterns effectively by introducing randomness to enhance model robustness.

Nature of the study:

The study "Crime Prediction Using Data Mining and Machine Learning" explores how various machine learning algorithms, including the Random Tree algorithm, can be applied to predict crime occurrences based on historical crime data. It analyzes crime-related features such as location, time, and type to classify and forecast potential crime events. By introducing randomness in feature selection at each split, the Random Tree algorithm helps build robust predictive models that reduce overfitting and improve accuracy. The study compares the performance of different classifiers and demonstrates the effectiveness of tree-based models in crime pattern prediction (Wu et al., 2019).

Relevance to our study:

This study is highly relevant to our project since both involve predicting binary outcomes — in our case, whether a crime case is "Solved" or "Unresolved." Like their crime prediction problem, our dataset contains structured categorical and numerical features related to cases. The demonstrated success of the Random Tree algorithm in analyzing crime data and making accurate predictions supports its suitability for our task. Using a similar approach allows us to leverage a proven method that balances interpretability and prediction accuracy in crime-related classification problems (Wu et al., 2019).

Random Tree - Use Training Set

```
Time taken to build model: 0.04 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.04 seconds

=== Summary ===

Correctly Classified Instances        1882               85.8577 %
Incorrectly Classified Instances       310               14.1423 %
Kappa statistic                          0.7172
Mean absolute error                      0.1888
Root mean squared error                  0.3073
Relative absolute error                 37.7672 %
Root relative squared error             61.4551 %
Total Number of Instances             2192

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.851    0.134    0.864      0.851    0.858      0.717    0.943     0.948     Solved
                 0.866    0.149    0.853      0.866    0.860      0.717    0.943     0.937     Unresolved
Weighted Avg.    0.859    0.141    0.859      0.859    0.859      0.717    0.943     0.942

=== Confusion Matrix ===

   a   b   <-- classified as
 933 163 |   a = Solved
 147 949 |   b = Unresolved
```

Random Tree - Ruleset

```
Classifier output
=== Classifier model (full training set) ===


RandomTree
==========

incidentTypeSpecific_Categorized = Armed_Encounter
|   SuspectKilled = No
|   |   offenseType = Batas_Pambansa : Solved (0/0)
|   |   offenseType = Crimes_Against_Chastity : Solved (0/0)
|   |   offenseType = Crimes_Against_Honor : Solved (0/0)
|   |   offenseType = Crimes_Against_Personal_Liberty_And_Security : Solved (0/0)
|   |   offenseType = Crimes_Against_Persons
|   |   |   typeofPlace_Categorized = Agricultural : Solved (0/0)
|   |   |   typeofPlace_Categorized = Commercial : Unresolved (1/0)
|   |   |   typeofPlace_Categorized = Industrial : Solved (0/0)
|   |   |   typeofPlace_Categorized = Institutional : Solved (0/0)
|   |   |   typeofPlace_Categorized = Parks_&_Recreation : Solved (0/0)
|   |   |   typeofPlace_Categorized = Protected_Area : Solved (0/0)
|   |   |   typeofPlace_Categorized = Residential
|   |   |   |   SuspectOccupation = Armed_Forces_Occupations : Solved (0/0)
|   |   |   |   SuspectOccupation = Clerical_Support_Workers : Solved (0/0)
|   |   |   |   SuspectOccupation = Craft_and_Related_Trades_Workers : Unresolved (1/0)
|   |   |   |   SuspectOccupation = Elementary_Occupations : Unresolved (1/0)
|   |   |   |   SuspectOccupation = Manager : Solved (0/0)
|   |   |   |   SuspectOccupation = Plant_and_Machine_Operators_and_Assemblers
|   |   |   |   |   VictimGender = Female : Unresolved (1/0)
|   |   |   |   |   VictimGender = Male : Solved (1/0)
|   |   |   |   SuspectOccupation = Professionals : Solved (0/0)
|   |   |   |   SuspectOccupation = Service_and_Sales_Workers : Solved (0/0)
|   |   |   |   SuspectOccupation = Skilled_Agriculture_Forestry_and_Fishery_Workers : Solved (2/0)
```

Explanation: The Random Tree model trained with the training set shows strong classification with an 85.86% success rate in classifying the 2,192 instances. It gets a Kappa statistic score of 0.7172, which indicates an average agreement between predicted values and actual values and hence a reliable performance. Key error metrics such as the mean absolute error (0.1888) and root mean squared error (0.3073) are both low, which is an indication of the reasonably good accuracy of the model in making predictions. Detailed accuracy by class, both "Solved" and "Unresolved" have shown high balanced performance with around 0.86 precision, recall, and F-measure values. This is also supported by the ROC and PRC areas greater than 0.93, confirming that the model does very well distinguishing classes. The model performs well on training data, but additional testing on unseen data would help evaluate overfitting as well as generalization. The confusion matrix further reinforces this performance, showing that the model accurately predicted 533 instances of “Solved” and 949 of “Unresolved” while misclassifying 163 and 147, respectively highlighting that although performance is strong, some room for improvement in minimizing false positives and negatives remains.

Random Tree - Supplied Test Set

```
Time taken to build model: 0.04 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Correctly Classified Instances          79               79      %
Incorrectly Classified Instances        21               21      %
Kappa statistic                          0.58
Mean absolute error                      0.2566
Root mean squared error                  0.4047
Relative absolute error                 51.3117 %
Root relative squared error             80.9317 %
Total Number of Instances              100

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.740    0.160    0.822      0.740    0.779      0.583    0.834     0.829     Solved
                 0.840    0.260    0.764      0.840    0.800      0.583    0.834     0.800     Unresolved
Weighted Avg.    0.790    0.210    0.793      0.790    0.789      0.583    0.834     0.814

=== Confusion Matrix ===

  a  b   <-- classified as
 37 13 |  a = Solved
  8 42 |  b = Unresolved
```

Explanation: Based on the above evaluation, the Random Tree model applied on the supplied test set achieved a classification accuracy value of 79 percent, which amounts to correctly classifying 79 out of the total 100 instances, with 21 instances (or 21 percent) being misclassified. Kappa statistic of 0.58 states that there is a moderate amount of agreement present among actual and predicted classes. Furthermore, errors were relatively quite low; with measures such as mean absolute error of 0.2566, root mean square error of 0.4047, and relative absolute error of 51.31 percent while root relative square error was 80.93%. For the class "Solved", model achieved precision of 0.822, recall value of 0.740 and F-measure of 0.779 while for the class of "Unresolved", it produced a score of 0.764 for precision, 0.840 recall, and 0.800 F-measure. According to the confusion matrix, it classified 37 "Solved" and 42 "Unresolved" instances correctly; however, there were slight misclassifications more in favor of the "Unresolved" class. Altogether, the model presents quite balanced performances with a bit of edge in recall for those unresolved issues.

Random Tree - 10 Folds Cross-validation

```
Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1689               77.0529 %
Incorrectly Classified Instances       503               22.9471 %
Kappa statistic                          0.5411
Mean absolute error                      0.2705
Root mean squared error                  0.4156
Relative absolute error                 54.098  %
Root relative squared error             83.129  %
Total Number of Instances             2192

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.771    0.230    0.770      0.771    0.771      0.541    0.824     0.818     Solved
                 0.770    0.229    0.771      0.770    0.770      0.541    0.824     0.766     Unresolved
Weighted Avg.    0.771    0.229    0.771      0.771    0.771      0.541    0.824     0.792

=== Confusion Matrix ===

   a   b   <-- classified as
 845 251 |   a = Solved
 252 844 |   b = Unresolved
```

Explanation: The Random Tree model with 10-fold cross-validation shows slightly lower performance than training set evaluation, as expected due to harsher validation. It classified 77.05% of cases correctly, identifying 1,689 of the 2,192 instances correctly. The Kappa statistic fell to 0.5411, indicating a moderate degree of agreement and possible overfitting in the training stage. Mean absolute error and root mean square error had risen to 0.2705 and 0.4156, respectively, indicating a loss in accuracy. Class-specific metrics, however, remained unchanged and ranging, with a precision, recall, and F-measure of 0.771 for both the "Solved" and "Unresolved" classes. The confusion matrix indicates, out of 845 "Solved" instances and 844 "Unresolved" instances correctly classified, 251 and 252 were incorrectly classified, respectively. The uniformity of citations shows that the model is somewhat balanced in differentiating classes but weak on generalization.

Random Tree - 80% Percentage Split

```
Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         331               75.5708 %
Incorrectly Classified Instances       107               24.4292 %
Kappa statistic                          0.5131
Mean absolute error                      0.2906
Root mean squared error                  0.4378
Relative absolute error                 58.1049 %
Root relative squared error             87.5279 %
Total Number of Instances              438

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.700    0.185    0.803      0.700    0.748      0.518    0.796     0.808     Solved
                 0.815    0.300    0.717      0.815    0.763      0.518    0.796     0.715     Unresolved
Weighted Avg.    0.756    0.240    0.761      0.756    0.755      0.518    0.796     0.763

=== Confusion Matrix ===

   a   b   <-- classified as
 159  68 |   a = Solved
  39 172 |   b = Unresolved
```

Explanation: The Random Tree model evaluated using an 80% percentage split achieved a classification accuracy of 75.57%, correctly classifying 331 out of 438 instances, while 107 instances (24.43%) were incorrectly classified. The Kappa statistic of 0.5131 indicates a moderate agreement between predicted and actual classifications. Error metrics show a mean absolute error of 0.2906, a root mean squared error of 0.4378, a relative absolute error of 58.10%, and a root relative squared error of 87.53%, suggesting moderate prediction deviations. For the “Solved” class, the model reached a precision of 0.803, recall of 0.700, and F-measure of 0.748, while the “Unresolved” class showed slightly lower recall at 0.815, with a precision of 0.717 and F-measure of 0.763. The weighted average F-measure was 0.755, indicating balanced performance, although the model slightly favors the “Unresolved” class in terms of precision. The confusion matrix indicates 159 "Solved" instances and 172 "Unresolved" instances were correctly predicted, while 68 "Unresolved" cases were incorrectly labeled as "Solved" and 39 "Solved" cases were mislabeled as "Unresolved," reflecting a slight tendency to favor the "Solved" class.

2. J48

(1) literature that used j48: **“Crime Prediction Using Decision Tree (J48) Classification Algorithm”**

The decision tree or j48 is one of the predictive modeling approaches used in statistics, data mining and machine learning. In data mining, decision trees can be described also as the combination of computational and mathematical techniques to aid the categorization, generalization of a given set of data.

Nature of the study:

A research study from Ahishakiye et al. (2017) was designed to address the challenges of effectively predicting crime categories to improve allocating law enforcement. By utilizing j48 algorithm, the authors developed a prototype model that analyzes crime data to forecast the likelihood of crimes occurred in various area. The study found that j48 algorithm predicted the unknown category of crime data by having the highest accuracy results, which is fair enough for the system to be relied for the prediction of crimes.

Relevance to our study:

This study gives relevance to our study as it shows how machine learning, especially the j48 decision tree algorithm can be utilized for predicting crime-related outcomes by analyzing historical data. In line with this, the methodology directly supports our objective of predicting whether a crime will be solved or remain unresolved. The relevance of this study suggests that it is practical and reliable model for crime prediction task by utilizing j48 algorithm that focuses on crime resolution outcomes (Bharadwaj et al., 2017).

(2) literature that used j48: **"Classification of Criminal Data Using J48-Decision Tree Algorithm"**

The J48 or Decision Tree algorithm is a simple yet widely used classification techniques that has a flowchart like tree structure. It works by recursively splitting data into subsets based on attribute values that provide the highest information gain.

Nature of the study:

From the study "Classification of Criminal Data Using J48 Algorithm" was address the large volumes of criminal data by analyzing and categorizing them to assist law enforcement to understand the patterns of crime and identify potential suspects. The author of this study utilized J48 decision tree algorithm due to its ability to handle categorical data effectively and generate classification model.

Relevance to our study:

This study is highly relevant to our research that aims to predict whether a crime will be solved or unresolved. By utilizing the j48 algorithm in their study it demonstrates how the algorithm handles categorical crime attributes and develops a clear decision rule, which similarly can apply to classify crime resolution status. Hence, j48 algorithm makes a valuable algorithm for predictive policing applications, which included to our study's focus on case outcome prediction (Kumari &Singh, 2014).

J48 - Use Training Set

```
Time taken to test model on training data: 0.03 seconds

=== Summary ===

Correctly Classified Instances        1762               80.3832 %
Incorrectly Classified Instances       430               19.6168 %
Kappa statistic                          0.6077
Mean absolute error                      0.2911
Root mean squared error                  0.3815
Relative absolute error                 58.2164 %
Root relative squared error             76.2997 %
Total Number of Instances             2192

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.779    0.172    0.820      0.779    0.799      0.608    0.857     0.853     Solved
                 0.828    0.221    0.790      0.828    0.809      0.608    0.857     0.813     Unresolved
Weighted Avg.    0.804    0.196    0.805      0.804    0.804      0.608    0.857     0.833

=== Confusion Matrix ===

   a   b   <-- classified as
 854 242 |   a = Solved
 188 908 |   b = Unresolved
```

Explanation: The J48 model with Use Training Set achieved highest results of accuracy with an 80.38%, correctly predicting 1,762 out of 2, 192 instances among the test options cross validation, and percentage spit. The model's shows 0.6077 of Kappa Statistic which indicates strong agreement and suggesting that the predictions are reliable. The ROC Area of 0.857 indicates for both classes an excellent ability to distinguish them. In addition, the evaluation matrix of both classes "Solved" and "Unresolved" shows well-balanced, with F-measure of 0.799 and 0.809 with a weighted average of 0.804. Hence, the J48 model with the use of training set illustrates good performance. Evaluating the performance, the model showed a recall of 0.779, precision of 0.820, and F-measure of 0.809 for the "Solved" Class. Oppositely, the "Unresolved" class, obtained precision of 0.790, recall was higher at 0.828, and the F-measure was 0.809, indicates slightly better in identifying unresolved cases. Additionally, the confusion matrix presents for the actual "Solved" instances that 854 were correctly classified, while misclassified were 188. In contrast, 908 instances of the actual "Unresolved" were correctly predicted, while 242 misclassified as "Solved".

J48 - Supplied Test Set

```
Time taken to build model: 0.07 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Correctly Classified Instances          74               74      %
Incorrectly Classified Instances        26               26      %
Kappa statistic                          0.48
Mean absolute error                      0.3158
Root mean squared error                  0.4192
Relative absolute error                 63.1528 %
Root relative squared error             83.8336 %
Total Number of Instances              100

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.680    0.200    0.773      0.680    0.723      0.483    0.801     0.821     Solved
                 0.800    0.320    0.714      0.800    0.755      0.483    0.801     0.742     Unresolved
Weighted Avg.    0.740    0.260    0.744      0.740    0.739      0.483    0.801     0.782

=== Confusion Matrix ===

  a  b   <-- classified as
 34 16 |  a = Solved
 10 40 |  b = Unresolved
```

Explanation: The evaluation of the J48 model using a supplied test set achieved a classification accuracy of 74%, with 74 out of 100 instances correctly classified and 26 instances (26%) misclassified. The model's Kappa statistic of 0.48 indicates a moderate level of agreement between the predicted and actual class labels beyond chance. The model showed a mean absolute error of 0.3158 and a root mean squared error of 0.4192, which suggests moderate prediction error. In addition, the relative absolute error stood at 63.15%, and the root relative squared error at 83.83%, reflecting a fair deviation from optimal predictions. Evaluating the performance, the model showed a recall of 0.680, precision of 0.773, and F-measure of 0.723 for the "Solved" Class. Conversely, the "Unresolved" class, obtained precision of 0.714, recall was higher at 0.800, and the F-measure was 0.755, indicates strong sensitivity in predicting unresolved cases. Furthermore, the confusion matrix presents for the actual "Solved" instances that 34 were correctly classified, while misclassified were 16. In contrast, 40 instances of the actual "Unresolved" were correctly predicted, while 10 misclassified as "Solved".

J48 - 10 Folds Cross-validation

```
Time taken to build model: 0.04 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1722               78.5584 %
Incorrectly Classified Instances       470               21.4416 %
Kappa statistic                          0.5712
Mean absolute error                      0.3068
Root mean squared error                  0.3996
Relative absolute error                 61.364  %
Root relative squared error             79.9193 %
Total Number of Instances             2192

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.755    0.184    0.804      0.755    0.779      0.572    0.825     0.819     Solved
                 0.816    0.245    0.769      0.816    0.792      0.572    0.825     0.778     Unresolved
Weighted Avg.    0.786    0.214    0.787      0.786    0.785      0.572    0.825     0.798

=== Confusion Matrix ===

   a   b   <-- classified as
 828 268 |   a = Solved
 202 894 |   b = Unresolved
```

Explanation: Given the evaluation result of J48 mode using 10-fold cross-validation achieved a classification accuracy of 78.56%, with 1,722 correctly predicting out of 2,192 instances, while 470 instances (21.44%) were misclassified. The model's Kappa statistic of 0.5712 indicates a moderate agreement between predicted and actual class labels beyond random chance. The model presents the mean absolute error of 0.3068 and a root mean squared error of 0.3996, resulting in moderate level of prediction error. Evaluating the performance, the model showed a recall of 0.755, precision of 0.804, and F-measure of 0.779 for the "Solved" Class. On the other hand, the "Unresolved" class, obtained precision of 0.769, recall was higher at 0.816, and the F-measure was 0.792, indicates strong sensitivity in predicting unresolved cases. Furthermore, the confusion matrix presents for the actual "Solved" instances that 828 were correctly classified, while misclassified were 202. In contrast, 894 instances of the actual "Unresolved" were correctly predicted, while 268 misclassified as "Solved".

J48 - 80% Percentage Split

```
Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         326               74.4292 %
Incorrectly Classified Instances       112               25.5708 %
Kappa statistic                          0.4901
Mean absolute error                      0.3271
Root mean squared error                  0.426
Relative absolute error                 65.3984 %
Root relative squared error             85.173  %
Total Number of Instances              438

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.696    0.204    0.786      0.696    0.738      0.494    0.797     0.802     Solved
                 0.796    0.304    0.709      0.796    0.750      0.494    0.797     0.729     Unresolved
Weighted Avg.    0.744    0.252    0.749      0.744    0.744      0.494    0.797     0.767

=== Confusion Matrix ===

   a   b   <-- classified as
 158  69 |   a = Solved
  43 168 |   b = Unresolved
```

Explanation: The evaluation of J48 model using 80% percentage split reached a classification accuracy of 74.43%, correctly predicting 326 out of 438 instances, and 112 instances (25.57%) were misclassified. The model's Kappa statistic of 0.4901 suggests a moderate agreement between the actual labels and the value beyond random chance. The model showed a mean absolute error of 0.3271 and root mean squared error of 0.426, which resulting in moderate prediction error. Evaluating the performance, the model showed a recall of 0.696, precision of 0.786, and F-measure of 0.738 for the "Solved" Class. On the other hand, the "Unresolved" class, obtained precision of 0.709, recall was higher at 0.796, and the F-measure was 0.750, suggest a better sensitivity in identifying unresolved cases. In addition, the confusion matrix presents for the actual "Solved" instances that 158 were correctly classified, while misclassified were 43. In contrast, 168 instances of the actual "Unresolved" were correctly predicted, while 69 misclassified as "Solved".

3. Random Forest

(1) literature that used Random Forest: **"CRIME RATE PREDICTION USING THE RANDOM FOREST ALGORITHM"**

In this study, Abdulraheem et al. (2022) employed the Random Forest algorithm to predict crime rates in Nigeria. The research aimed to identify the frequency and patterns of various crimes, including cybercrime, bribery, corruption, robbery, and money laundering. By processing and visualizing raw datasets, the study utilized machine learning methods to extract knowledge and uncover hidden patterns in the data. These patterns were then used to investigate and report on crime trends, providing valuable insights for crime analysts and aiding in crime prevention efforts.

Nature of the study:

The research conducted by Abdulraheem et al. (2022) focused on applying machine learning algorithms, particularly Random Forest, to anticipate crime distribution over large areas in Nigeria. The study involved processing raw datasets and employing various visualization techniques to understand crime patterns. The machine learning methods deployed aimed to extract knowledge from extensive datasets, uncover hidden patterns, and utilize these insights to investigate and report on crime trends. The ultimate goal was to support law enforcement agencies in making informed decisions and implementing preventative measures against crime.

Relevance to our study

This research aligns with our study's objective of forecasting crime resolution outcomes using machine learning algorithms. Similar to Abdulraheem et al. (2022), our study aims to utilize advanced algorithms to predict whether a case is solved or unsolved. The use of Random Forest in their study demonstrates the algorithm's effectiveness in handling complex datasets and uncovering significant patterns related to crime occurrences. By drawing parallels with their methodology, we can enhance our approach to crime prediction and improve the accuracy of our forecasting models.

(2) literature that used ZeroR: **"Prediction of Crime Hotspots based on Spatial Factors of Random Forest"**

In this study, Yao et al. (2020) explored the application of the Random Forest algorithm to predict crime hotspots by analyzing spatial factors. The research aimed to identify regions with a high probability of crime occurrence and visualize crime-prone areas. By leveraging machine learning techniques, particularly Random Forest, the study sought to enhance the accuracy of crime hotspot predictions and provide valuable insights for law enforcement agencies.

Nature of the study:

The research conducted by Yao et al. (2020) focused on utilizing spatial factors in conjunction with the Random Forest algorithm to predict crime hotspots. The study involved analyzing various spatial inputs and employing machine learning methods to forecast areas with a high likelihood of criminal activity. The objective was to develop a system capable of accurately predicting and visualizing regions prone to crime, thereby aiding in proactive policing and resource allocation.

Relevance to our study:

This research aligns with our study's objective of forecasting crime resolution outcomes using machine learning algorithms. Similar to Yao et al. (2020), our study aims to utilize advanced algorithms to predict whether a case is solved or unsolved. The use of Random Forest in their study demonstrates the algorithm's effectiveness in handling complex datasets and uncovering significant patterns related to crime occurrences. By drawing parallels with their methodology, we can enhance our approach to crime prediction and improve the accuracy of our forecasting models.

Random Forest - Use Training Set

```
Time taken to test model on training data: 0.16 seconds

=== Summary ===

Correctly Classified Instances        1882               85.8577 %
Incorrectly Classified Instances       310               14.1423 %
Kappa statistic                          0.7172
Mean absolute error                      0.2111
Root mean squared error                  0.3139
Relative absolute error                 42.2198 %
Root relative squared error             62.7748 %
Total Number of Instances             2192

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.832    0.115    0.879      0.832    0.855      0.718    0.938     0.947     Solved
                 0.885    0.168    0.841      0.885    0.862      0.718    0.938     0.930     Unresolved
Weighted Avg.    0.859    0.141    0.860      0.859    0.858      0.718    0.938     0.939

=== Confusion Matrix ===

   a   b   <-- classified as
 912 184 |   a = Solved
 126 970 |   b = Unresolved
```

Explanation: The evaluation of the Random Forest model using the training set achieved a high classification accuracy of 85.86%, correctly classifying 1,882 out of 2,192 instances, while 310 instances (14.14%) were misclassified. With a Kappa statistic of 0.7172, the model shows a strong agreement between predictions and actual labels beyond random chance. It also demonstrated low prediction errors, with a mean absolute error of 0.2111 and a root mean squared error of 0.3139. For class-specific performance, the “Solved” class attained a precision of 0.879, recall of 0.832, and F-measure of 0.855, while the “Unresolved” class showed slightly better recall at 0.885, with a precision of 0.841 and F-measure of 0.862. The confusion matrix indicates that 912 actual “Solved” instances were correctly predicted, while 184 were misclassified as “Unresolved”; meanwhile, 970 actual “Unresolved” instances were correctly classified, and 126 were misclassified as “Solved.” These results highlight the model’s strong and balanced performance in predicting both classes.

Random Forest - Supplied Test Set

```
Time taken to build model: 0.32 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Correctly Classified Instances          79               79      %
Incorrectly Classified Instances        21               21      %
Kappa statistic                          0.58
Mean absolute error                      0.2684
Root mean squared error                  0.3827
Relative absolute error                 53.6845 %
Root relative squared error             76.5344 %
Total Number of Instances              100

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.720    0.140    0.837      0.720    0.774      0.586    0.860     0.865     Solved
                 0.860    0.280    0.754      0.860    0.804      0.586    0.860     0.854     Unresolved
Weighted Avg.    0.790    0.210    0.796      0.790    0.789      0.586    0.860     0.859

=== Confusion Matrix ===

  a  b   <-- classified as
 36 14 |  a = Solved
  7 43 |  b = Unresolved
```

Explanation: The evaluation of the Random Forest model on the supplied test set yielded an overall classification accuracy of 79%, correctly identifying 79 out of 100 instances, while 21% were misclassified. A Kappa statistic of 0.58 indicates moderate agreement between the predicted and actual labels beyond chance. The model exhibited moderate prediction error, with a mean absolute error of 0.2684 and a root mean squared error of 0.3827. In terms of class-specific performance, the "Solved" class achieved a precision of 0.837, recall of 0.720, and F-measure of 0.774, whereas the "Unresolved" class performed slightly better with a recall of 0.860, precision of 0.754, and F-measure of 0.804. According to the confusion matrix, 36 "Solved" instances were correctly classified and 14 were misclassified, while 43 "Unresolved" instances were accurately predicted and 7 were misclassified, indicating the model is more effective at identifying unresolved cases.

Random Forest - 10-Fold Cross-validation

```
Time taken to test model on test split: 0.03 seconds

=== Summary ===

Correctly Classified Instances         334               76.2557 %
Incorrectly Classified Instances       104               23.7443 %
Kappa statistic                          0.5272
Mean absolute error                      0.2972
Root mean squared error                  0.4165
Relative absolute error                 59.4265 %
Root relative squared error             83.2676 %
Total Number of Instances              438

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.696    0.166    0.819      0.696    0.752      0.534    0.824     0.862     Solved
                 0.834    0.304    0.718      0.834    0.772      0.534    0.824     0.767     Unresolved
Weighted Avg.    0.763    0.232    0.770      0.763    0.762      0.534    0.824     0.816

=== Confusion Matrix ===

   a   b   <-- classified as
 158  69 |   a = Solved
  35 176 |   b = Unresolved
```

Explanation: The evaluation of the Random Forest model using 10-fold cross-validation achieved a classification accuracy of 76.26%, with 334 out of 438 instances correctly classified and 104 instances (23.74%) misclassified. The Kappa statistic of 0.5272 indicates moderate agreement beyond chance between actual and predicted values. The model's mean absolute error was 0.2972 and root mean squared error was 0.4165, suggesting a moderate level of prediction error. The "Solved" class achieved a precision of 0.819, recall of 0.696, and F-measure of 0.752, while the "Unresolved" class yielded a precision of 0.718, recall of 0.834, and F-measure of 0.772, indicating the model is slightly better at identifying unresolved cases. The confusion matrix shows that 158 "Solved" cases were correctly identified, while 69 were misclassified, and for the "Unresolved" category, 176 were correctly predicted, and 35 were misclassified as “Solved.” Overall, the model exhibits balanced and moderately strong performance across both classes.

Random Forest - 80% Percentage Split

```
Time taken to build model: 0.15 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.03 seconds

=== Summary ===

Correctly Classified Instances         336               76.7123 %
Incorrectly Classified Instances       102               23.2877 %
Kappa statistic                          0.5363
Mean absolute error                      0.2975
Root mean squared error                  0.4176
Relative absolute error                 59.4896 %
Root relative squared error             83.4984 %
Total Number of Instances              438

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.700    0.161    0.824      0.700    0.757      0.543    0.823     0.861     Solved
                 0.839    0.300    0.722      0.839    0.776      0.543    0.823     0.765     Unresolved
Weighted Avg.    0.767    0.228    0.775      0.767    0.766      0.543    0.823     0.815

=== Confusion Matrix ===

   a   b   <-- classified as
 159  68 |   a = Solved
  34 177 |   b = Unresolved
```

Explanation: The Random Forest model evaluated using an 80% percentage split achieved a classification accuracy of 76.71%, correctly classifying 336 out of 438 instances, while 102 instances (23.29%) were misclassified. The Kappa statistic of 0.5363 indicates a moderate level of agreement between predictions and actual outcomes beyond chance. Prediction error metrics included a mean absolute error of 0.2975 and a root mean squared error of 0.4176, suggesting moderate prediction precision. For class-specific performance, the "Solved" class achieved a precision of 0.824, recall of 0.700, and F-measure of 0.757, while the "Unresolved" class attained a higher recall of 0.839, with a precision of 0.722 and F-measure of 0.776. The confusion matrix shows that 159 "Solved" instances were correctly identified, while 68 were misclassified as "Unresolved"; conversely, 177 "Unresolved" instances were correctly classified, with 34 misclassified as "Solved," reflecting the model’s relatively stronger performance on unresolved cases.

Summary Result

	Accuracy	Kappa Value	Mean Absolute error	TP Rate	FP Rate	Precision	Recall	F-measure	Time to build
Random Tree - Use Training Set	85.8577%	0.7172	0.1888	0.859	0.141	0.859	0.859	0.859	0.04
Random Tree - Supplied Test Set	79%	0.58	0.2566	0.790	0.210	0.793	0.790	0.789	0.01
Random Tree - 10 Fold Cross-validation	77.0529%S	0.5411	0.2705	0.771	0.229	0.771	0.771	0.771	0.02
Random Tree - 80% Percentage Split	75.5708%	0.5131	0.2906	0.756	0.240	0.761	0.756	0.755	0
J48 - Use Training Set	80.3832%	0.6077	0.2911	0.804	0.196	0.805	0.804	0.804	0.03
J48 - Supplied Test Set	74%	0.48	0.3158	0.740	0.260	0.744	0.740	0.739	0.01
J48 - 10 Fold Cross-validation	78.5584%	0.5712	0.3068	0.786	0.214	0.787	0.786	0.785	0.04
J48 - 80% Percentage Split	74.4292%	0.4901	0.3271	0.744	0.252	0.749	0.744	0.744	0
Random Forest - Use Training Set	85.8577%	0.7172	0.2111	0.859	0.141	0.860	0.859	0.858	0.5
Random Forest - Supplied Test Set	79%	0.58	0.2684	0.790	0.210	0.796	0.790	0.789	0.32
Random Forest – 10 Fold Cross-validation	78.1934	0.5639	0.2751	0.782	0.218	0.782	0.782	0.782	0.26
Random Forest - 80% Percentage Split	76.2557	0.5272	0.2972	0.763	0.232	0.770	0.763	0.762	0.22

5. Reflection

The toughest hurdles give out better lessons and experiences in the long run. This project made me think, feel, and act as a business analyst student who also delved into the system development aspect, key to WMAD students. I can safely say that this project is the most difficult I've had so far in college. The endless data preprocessing, model building, checking, and system development motivated us to achieve more and aim higher.

But despite all those hardships, I can't help but feel a sense of relief and calmness knowing that I have these people by my side. This project wouldn't be possible without the full cooperation of my team, and I'm grateful to them. This project, this subject, and this class in general taught me so much that I know I can use for the betterment of my career in analytics. Chico, Ken

What this project left in me was the importance of trust and patience. The complicated cleaning of data, selection of attributes, and nearly endless clicking on Weka to find the best model. It sure was frustrating at times, especially when things didn't go the way we expected them to, and I wasn't sure if I was doing things correctly.

But trusting the process, and more importantly, trusting the members, made a big difference. It wasn't just about getting things done. It was about working together, supporting each other, and combining our knowledge and strengths. I am well aware that I am not good at everything, but they helped fill in the gaps, and we balanced each other really well. Also, our skills were honed, and we are able to generate promising insights, in my opinion. It was challenging but, overall, really rewarding. De Jesus, Princess Anne

"Do not get overwhelmed." This seemingly simple phrase became my unexpected lifeline during this project. The initial stages were incredibly challenging—the complexity of the data, the pressure to meet deadlines, and the constant need for precision created a significant sense of anxiety. However, repeatedly reminding myself of this motto helped me re-center. It encouraged a more methodical approach, prompting me to pause, breathe, and strategize rather than panicking. It transformed a potentially paralyzing experience into a manageable, even rewarding, one. This project emphasized the challenges of dealing with real data, requiring strict justification of each step and an ongoing awareness of potential bias, counting heavily on statistical support and research. This project also pointed out the value of each person played bringing his or her skills, ideas, and knowledge, finding common ground between a diverse set of opinions and understanding presented a large obstacle, but the combined efforts produced results we all aim. Difficult as it was, the journey yielded invaluable learning experiences, with both successes and failures serving as teachers. Lastly, I extend my sincere gratitude to our professors for their invaluable guidance and to my amazing groupmates, whose made this challenging project both enjoyable and rewarding. Garces, Grace

Working on this project has been a fulfilling journey for me alongside my group. The trust and patience we had with one another made our teamwork strong and compatible. From finding our client to gathering the necessary resources, we all poured in hard work and dedication to ensure the success of this project. There were times when we faced challenges during the preprocessing stage, especially when we already had the model but unfortunately needed to clean our datasets due to some anomalies we discovered.

For the betterment and faster progress of this project, our group divided responsibilities based on each member's strengths — from system development to documentation. I took part in developing the system as the front-end developer, making sure it was responsive, visually appealing, and user-friendly, especially for older users. While each member focused on their assigned tasks, we didn't limit ourselves to our individual roles. We embraced shared responsibilities and continuously helped one another. This experience will always be a part of me as I continue my journey, both inside and outside of this team. I'm especially grateful to our professor, whose guidance and unwavering support played a key role in our growth—her advice truly made a lasting impact on us. Lachica, Crystel

This project has been a transformative experience, allowing me to gain valuable knowledge and improve both technical and non-technical skills. Throughout the process, we faced challenges and had to make important decisions. Some setbacks were frustrating, especially when things didn't turn out as expected. But having a reliable and dedicated team helped a lot—they kept me motivated, and we pushed through the difficulties together.

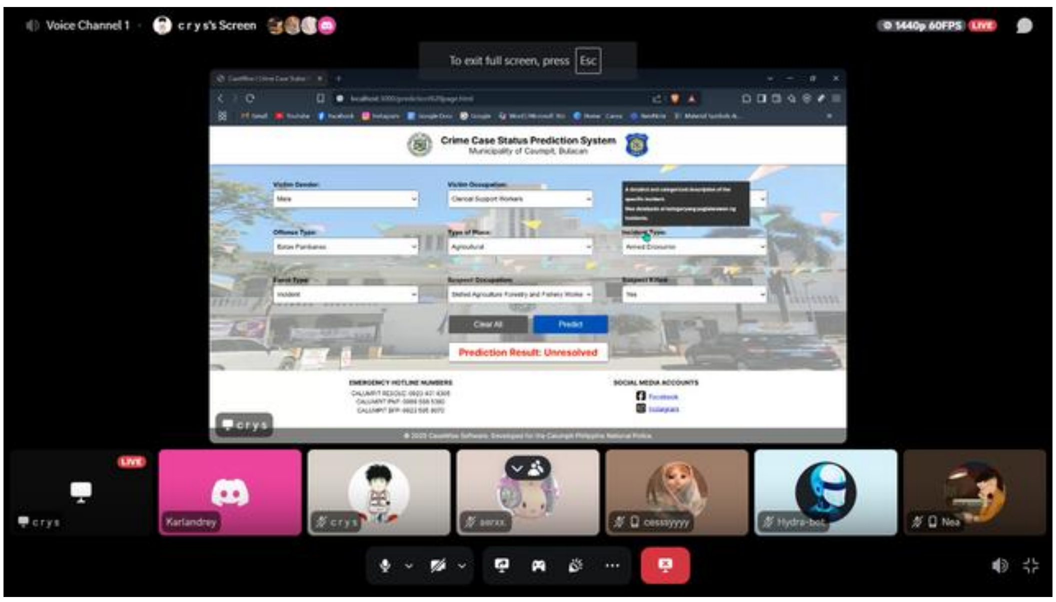
Analyzing data, refining attribute selection, and performing classification were key steps in making sure we got the best results for our system. Testing different algorithms to find the most effective one took patience, problem-solving, and adaptability. It wasn't easy but seeing progress every day made the hard work worth it. Besides learning technical skills, this project showed me how important it is to make informed decisions instead of focusing too much on uncertainties. Collaboration was also a big factor—working with committed teammates made our ideas come to life. I'm truly grateful to my group mates for their dedication and teamwork. Their effort and willingness to improve made this experience even more meaningful. Macabenta, Laarnie

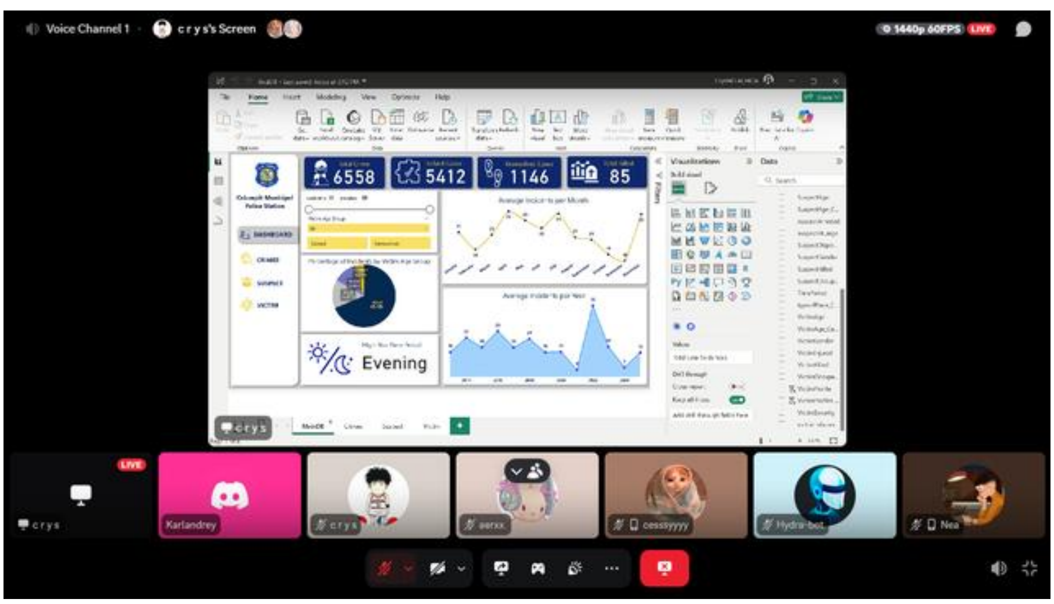
Throughout the process of this project, we spent countless sleepless nights just to ensure our data was clean and reliable. We went back and forth with our client several times to gain clarity and confidence in the direction we chose. We analyzed our data almost every single day, carefully deciding on the best predictions and visualizations to present. And there were moments when our patience was nearly exhausted from the continuous work. But despite all of that, I realized one important truth—things become more manageable when you're working with responsible teammates and when there's open and effective communication. That became our foundation, helping us better understand and appreciate the course and specialization we're pursuing.

For me, this project is more than just something to submit or present. It became a source of motivation—a turning point that pushed me to aim higher, improve myself, and embrace my role as a more responsible leader. Mendoza, Carl Andrei

GROUP PHOTO:


Voice Channel 1
Karlandrey's Screen
720p 30FPS
LIVE
Karlandrey


Voice Channel 1
c r y s's Screen
1440p 60FPS
LIVE
To exit full screen, press Esc
Crime Case Status Prediction System
Prediction Result: Unresolved


Voice Channel 1
c r y s's Screen
1440p 60FPS
LIVE
6558
5412
1146
85
Evening

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Part 2

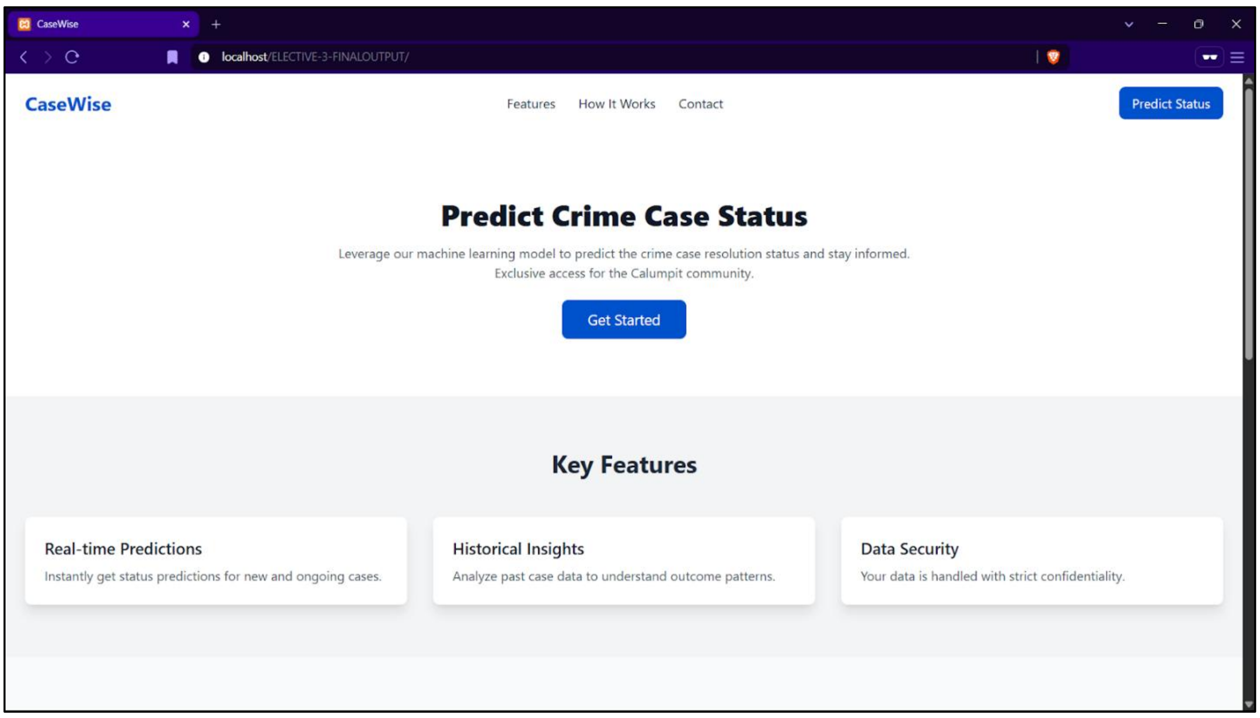
System Development

The system was created using HTML, CSS, and JavaScript. HTML served as the backbone of the project, it formed the structure of the whole webpage and the contents of the webpage. CSS was used to add a simple and minimalistic design and theme for the system, creating a stylistic, modern look and feel for the users. Lastly, JavaScript was used to handle the ruleset derived from WEKA, handling the main logic (the if structure) of the system.

1. System

Figure 14

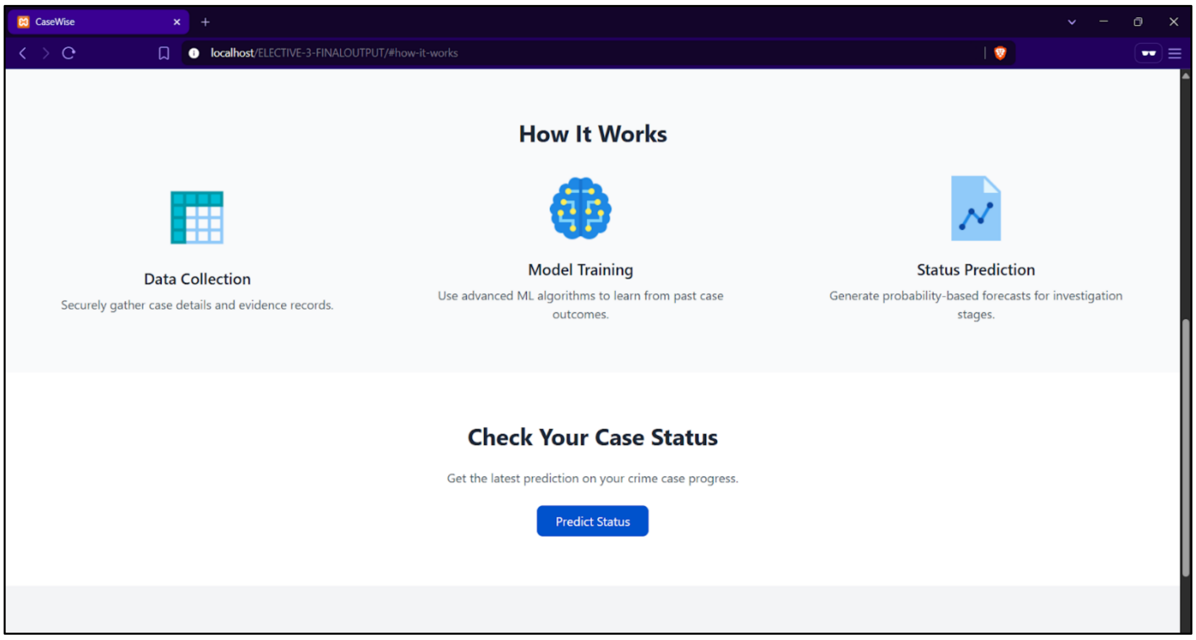
CaseWise – Landing Page and Features



The figure shows the landing page of CaseWise. It is a static landing page created using HTML and CSS. The user is greeted with a brief piece of information about what the system is used for, along with its key features. The header provides 4 navigation bars: Features, How it Works, Contact, and Predict Status. These 4 buttons will navigate the user to their corresponding sections on the same webpage.

Figure 15

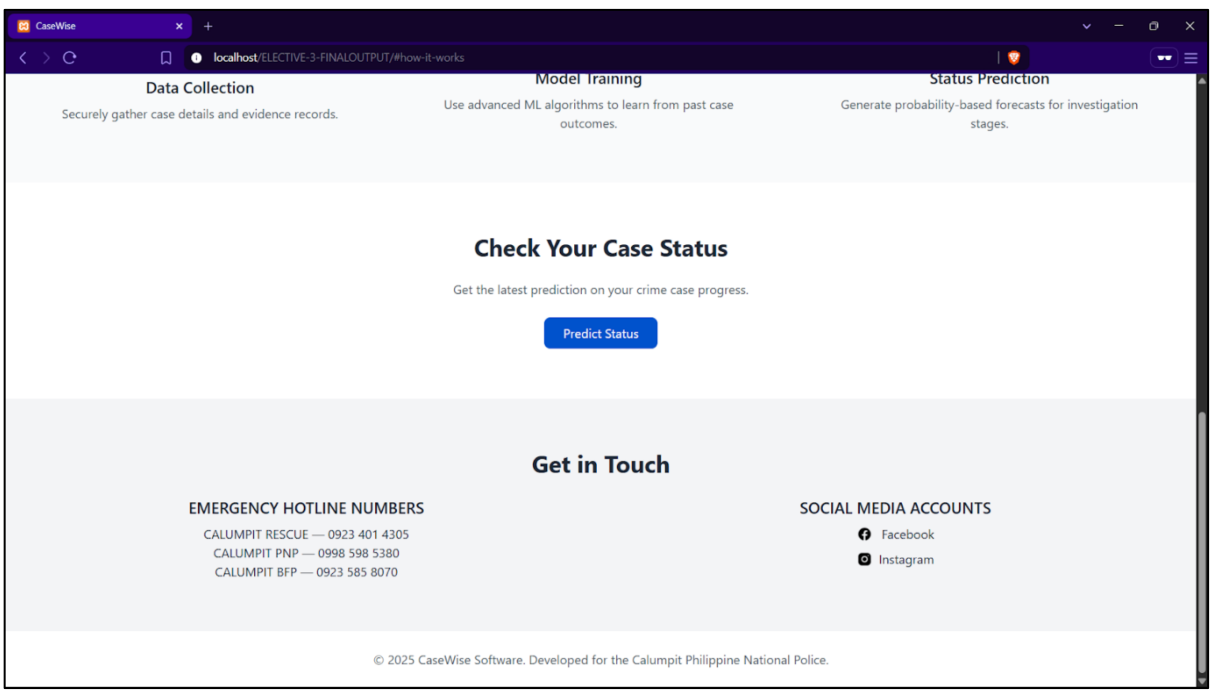
CaseWise – How it Works



The figure above shows the How it Works section of the landing page, telling the end user a brief introduction about how the system was created and how it generates its predictions. This helps the user get to understand the general context of how everything was pieced together, without the complicated technicalities that the user doesn't necessarily need to know. By clicking the Predict Status button, it redirects the user to the prediction page.

Figure 16

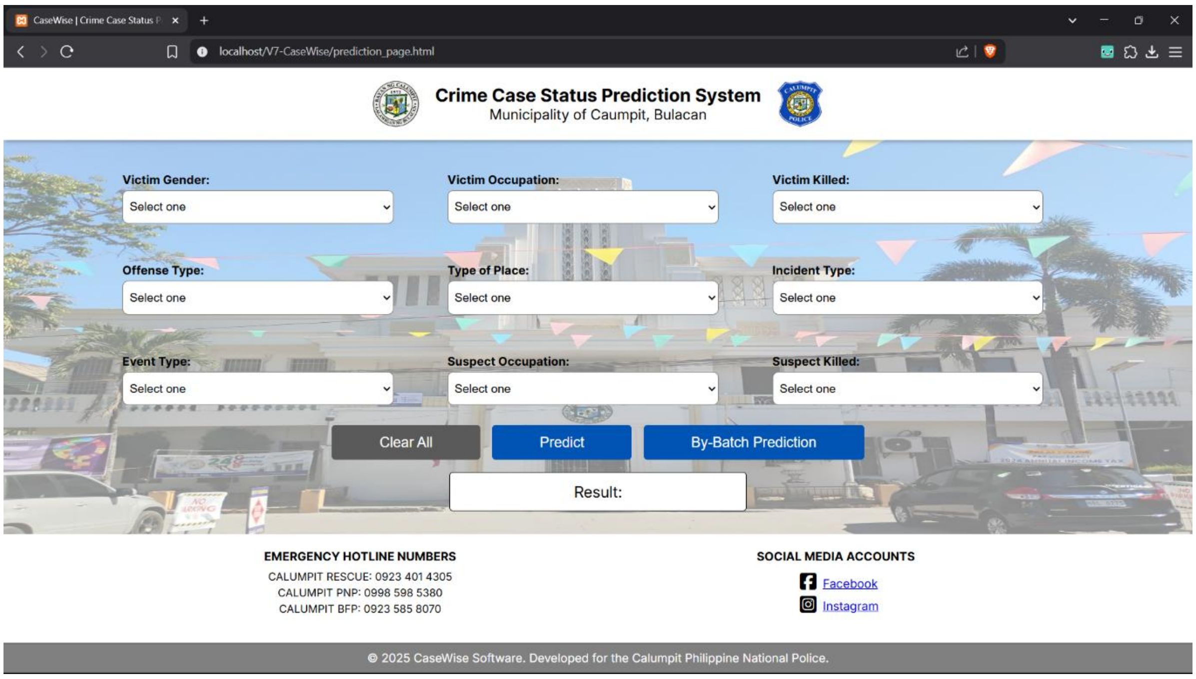
CaseWise – Contact



The figure above highlights the footer section of the webpage, containing important contacts for Calumpit Rescue, Calumpit Philippine National Police (PNP), and Calumpit Bureau of Fire Protection (BPF) along with the social media accounts of Calumpit Municipal Police Station (MPS) on Facebook and the Instagram account of Calumpit Fire Station.

Figure 17

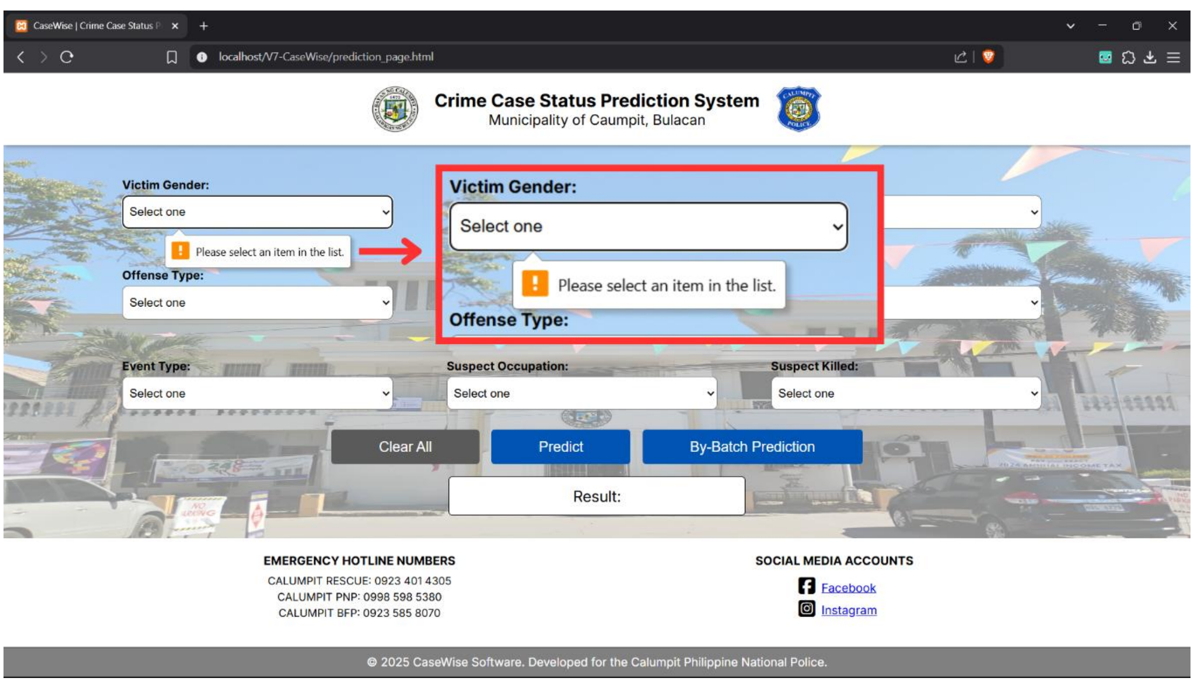
CaseWise – Prediction Page



The figure shows the prediction page of CaseWise. It contains a header to display the title of the system and to whom it is. It contains 9 drop-down lists to control the inputs of the user. To mitigate input error, the user must explicitly select the value from the drop-down list that is already defined in the system. Like the landing page, it also contains the same footer, containing necessary contact information and social media links. Clicking the Clear All button will refresh all the drop-down lists as well as the result container.

Figure 18

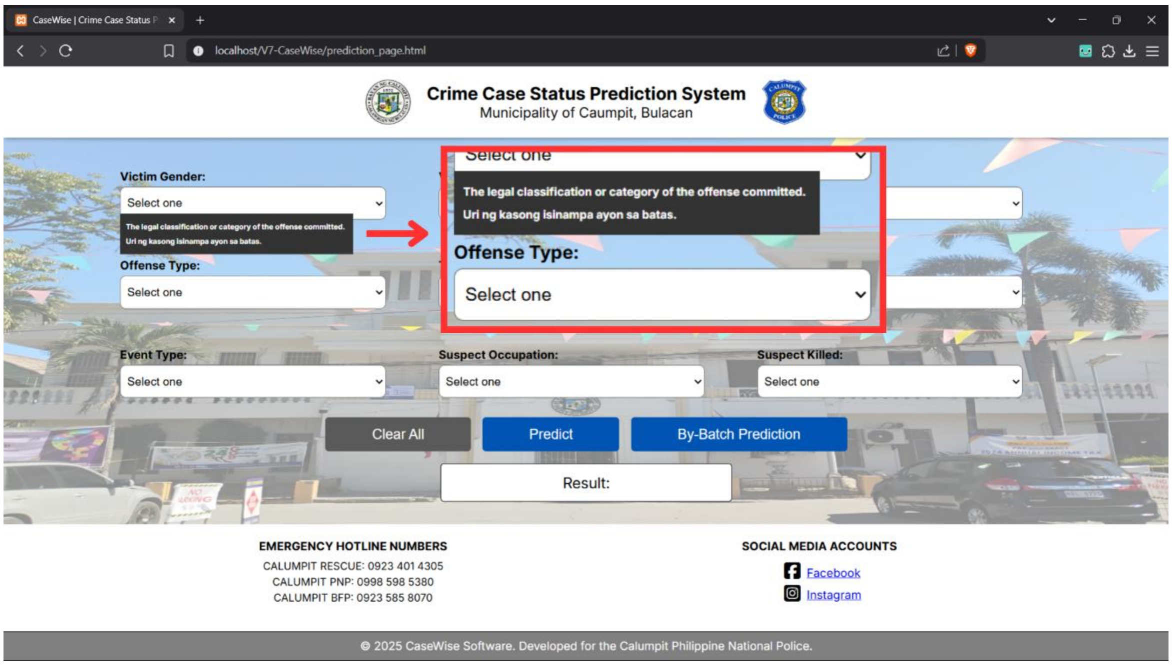
CaseWise – Prediction Page – Handling Blank Inputs



If the user fails to select a value from at least one of the 9 drop-down lists and the user clicks the Predict button, it will prompt a warning notification as shown in the red box in the figure above. Notifying the user to select proper values in the list.

Figure 19

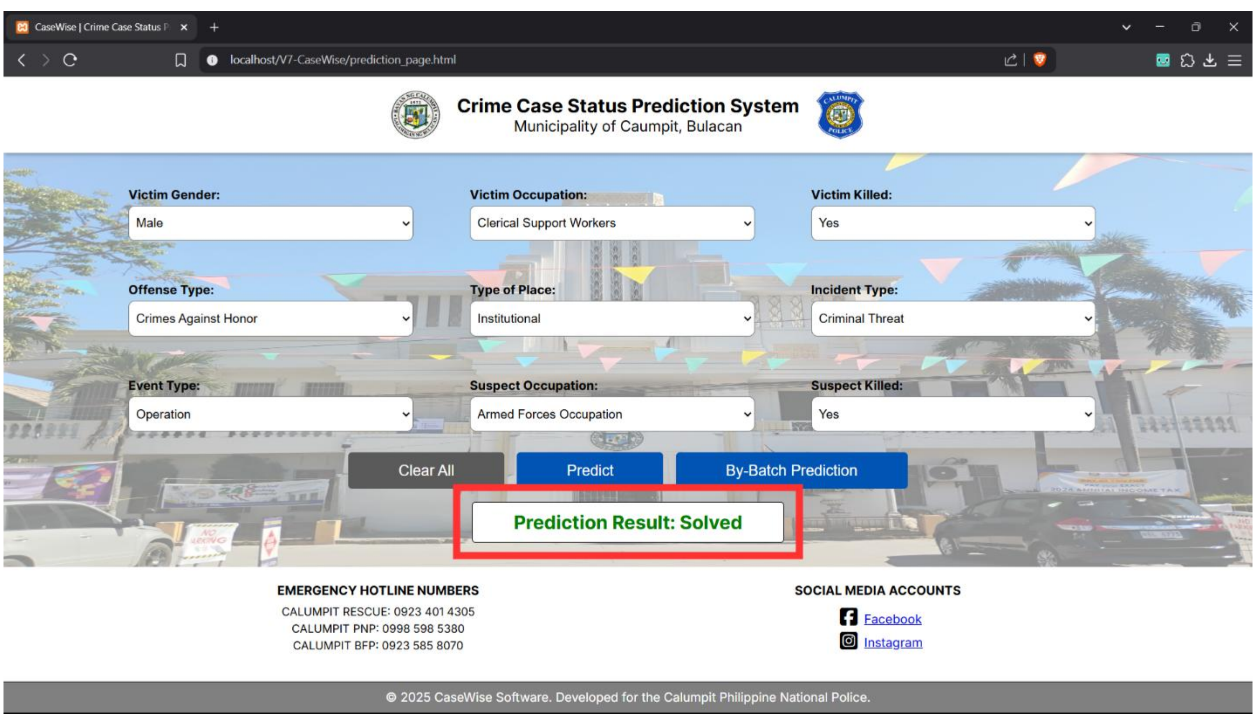
CaseWise – Prediction Page – Display Attribute Information on Hover



A short description about the attributes is displayed if the user hovers above their corresponding label, as shown in the figure above, highlighted in red, where the user hovered in the Offense Type label. It includes both English and Tagalog descriptions to guide the user.

Figure 20

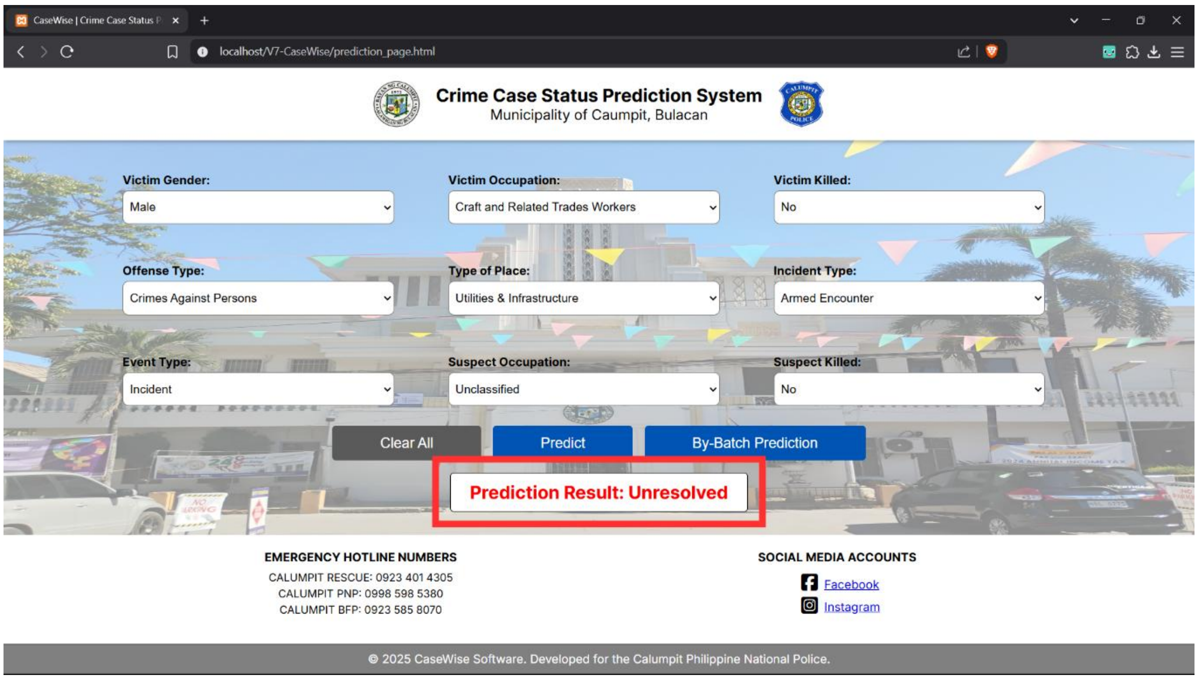
CaseWise – Prediction Page – Solved Output



The figure above shows the output of the system if the case that the user is trying to predict is going to be solved, highlighted in red. The text turns from black to green, signifying a positive result.

Figure 21

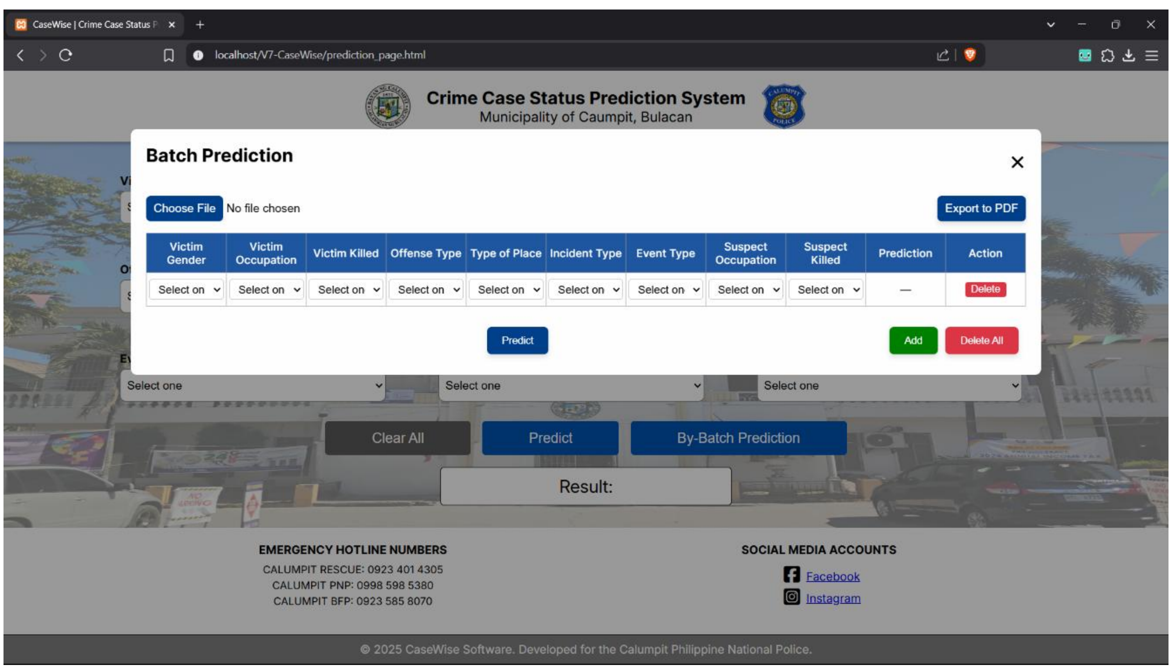
CaseWise – Prediction Page – Unresolved Output



The figure above shows the output of the system if the case that the user is trying to predict is not going to be solved, highlighted in red. The text turns from black to red, signifying a negative result.

Figure 22

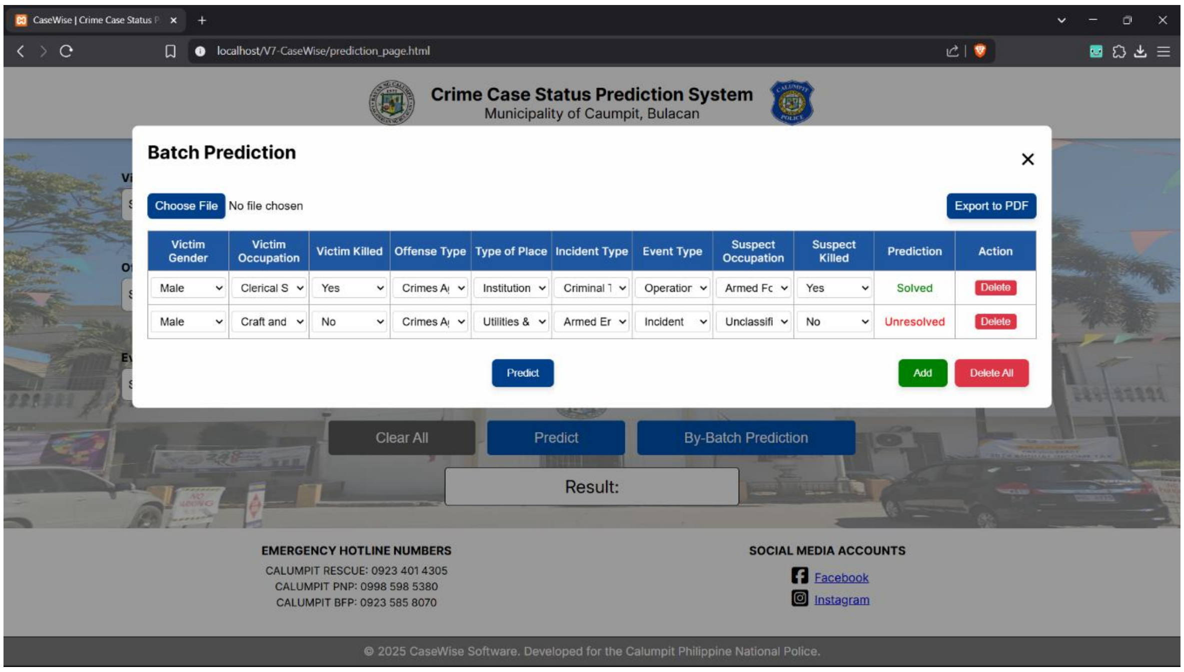
CaseWise – Prediction Page – Batch Prediction



Clicking the By-Batch Prediction button, the system will prompt a pop-up page where the user can easily add rows manually or upload a premade CSV/XSL of the rows that the user wants to predict. The pop-up contains all the variables and are also in order parallel to the prediction page.

Figure 23

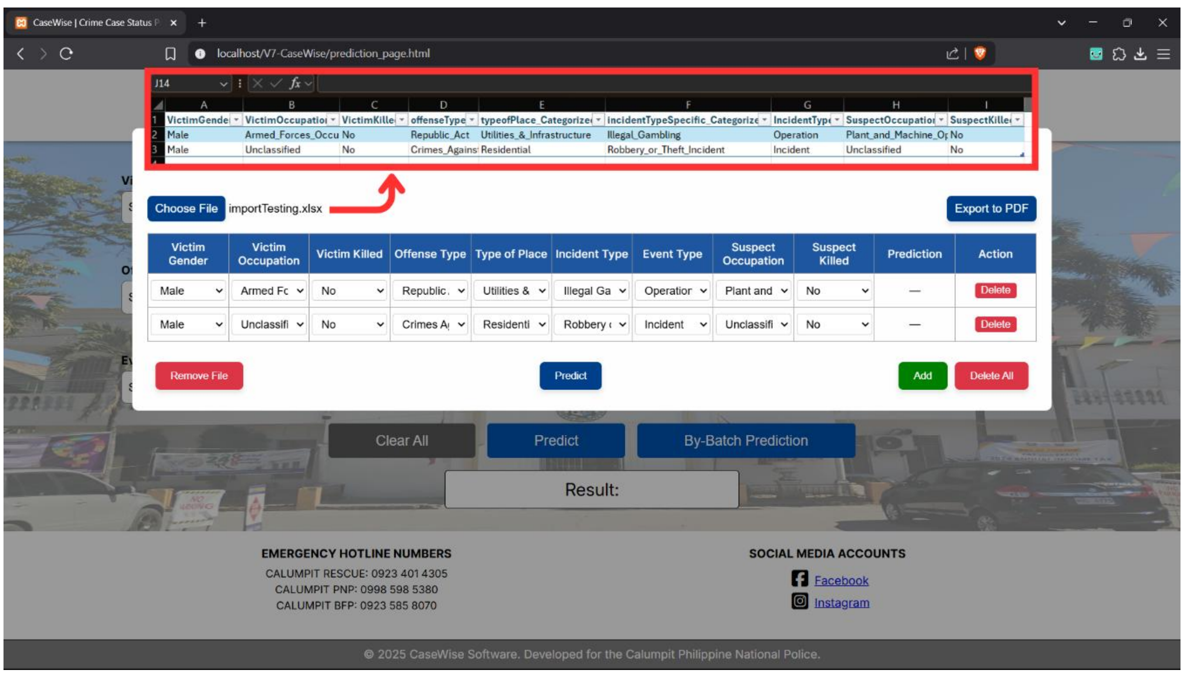
CaseWise – Prediction Page – Batch Prediction Results



The predict button in the pop-up works the same way as the predict button in the prediction page. Clicking the button will show all the results of the rows as shown in the figure above.

Figure 24

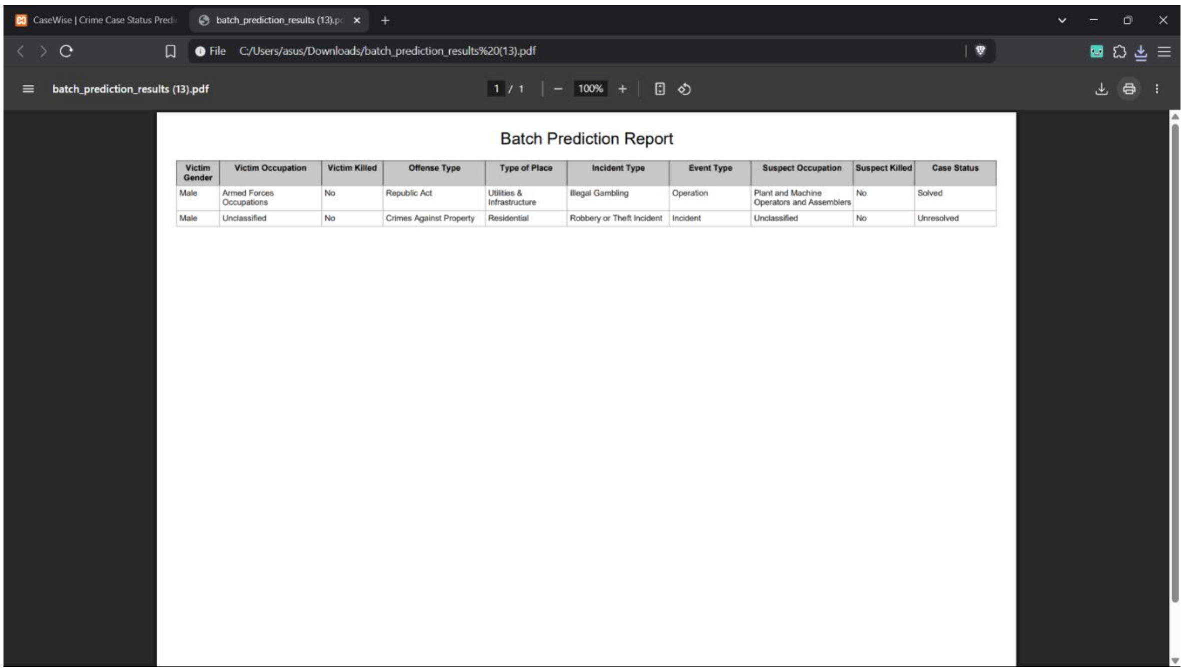
CaseWise – Prediction Page – Import XSL/CSV File



To even support predictions by bulk, CaseWise provides an option for the users to upload a premade XSL/CSV file for batch predictions. It's important to note that if the user wishes to upload a premade XSL/CSV file, they must strictly follow the format of the file as shown in the figure above. The format follows how the variables were laid out in the system. Following the format ensures consistency and accuracy of the variables inputted by the user. At the same time, the values of the variables in the system follows a title case format with underscores ("_") as spaces (i.e., "Illegal_Gambling"). The user must follow the strict format of the values if they wish to upload files for the system to avoid inaccuracy and input errors.

Figure 25

CaseWise – Export To PDF

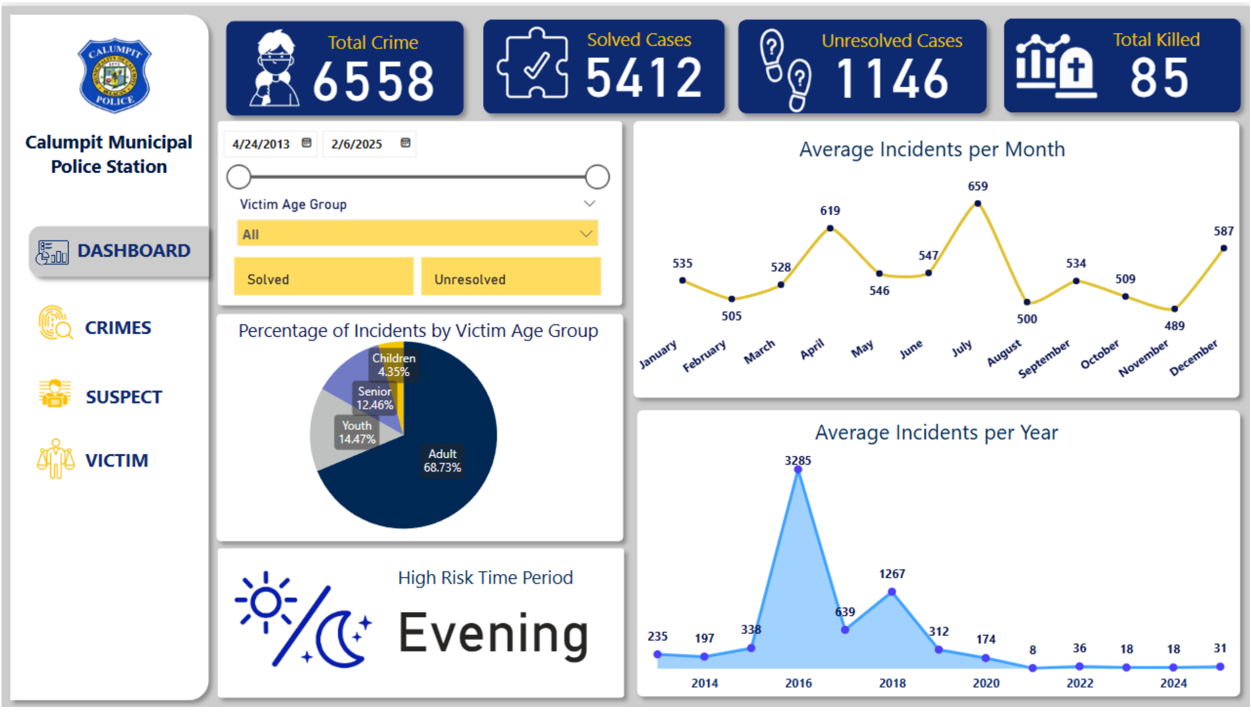


Clicking the export to PDF button, using the jsPDF library, it saves the output from the pop-up table to a PDF file for easy access and reference to the results. This is done using the jsPDF autoTable plugin to easily convert the HTML table from the pop-up to a structured table in PDF form.

2. Visual Analytics

Figure 26

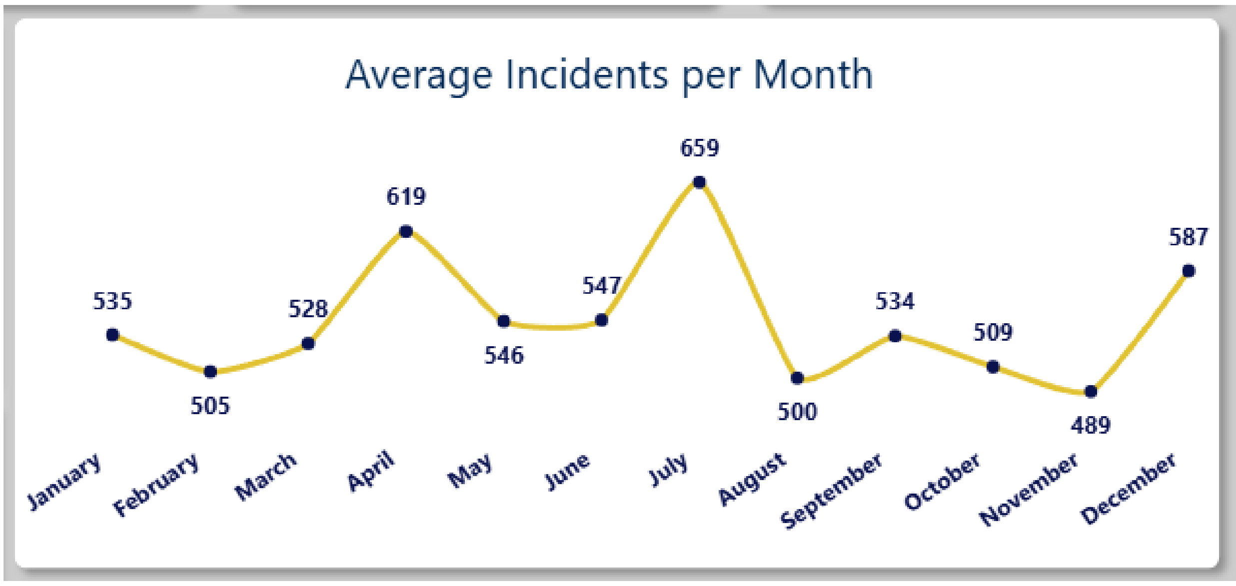
CMPS Dashboard Tab



The Dashboard Page of Calumpit Municipal Police Station presents the professional overview and comprehensive crime data from April 2013 to February 2025. The Card that shows the total crime, total number of solved cases, total number of unresolved cases, and total killed provides a clear value and the needs for continued focus on both investigative work and preventive measures. The slicer part indicates for the case status, victim age group, and the crime date committed, this allows the stakeholder to refine data based on specific timeframe and victim age demographics. The reported crimes achieve a total of 6,558, with the highest number of solved cases, about 5,412, which reflects an effective policing effort, through the presence of 1,146 unresolved cases and 85 total fatalities highlighting areas that require continuing attention. The percentage of Incidents by Victim Age Group shows that Adults were majority of the victims (68.73%), highlighting further targeted prevention strategies, while the incident peaks showed during the month of July and evening hours, which indicates the need for more deployment of police officer. Additionally, the average incident per year surge in 2016, emphasize the highest number of incidents happen during the mentioned year.

Figure 27

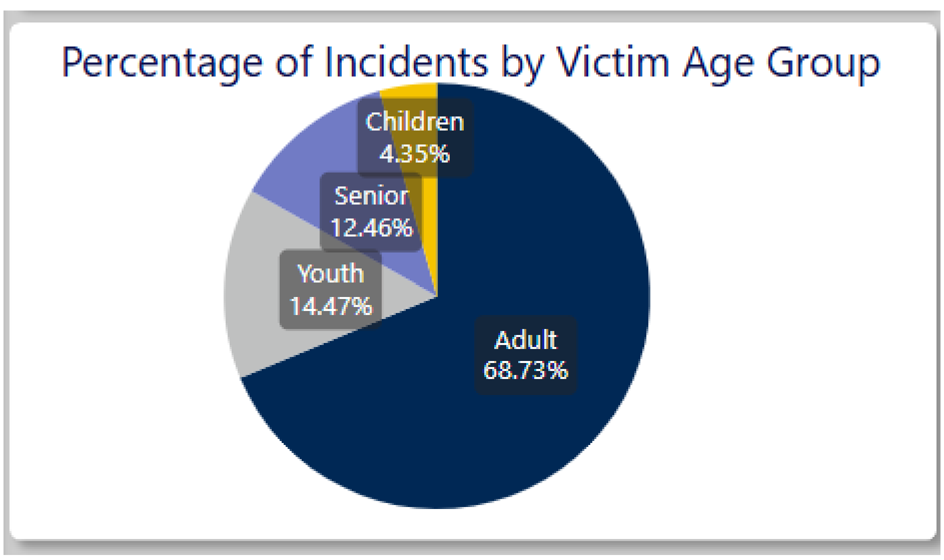
CMPS Dashboard - Average Incident Per Month



This line graph shows the calculated average number of incidents per month from the column incidentTypeSpecific_Categorized. With an average of 659 incidents, the month of July is revealed as the peak month which suggests possible event-related or seasonal spikes. It shows that in the month of November (489), it appears to be the lowest average of incidents from 2013 to 2025. In line with this, high authorities could guide resource allocation during high-risk month. Understanding these patterns is important for adjusting the patrol schedules and launching public safety campaigns to decrease the number of incidents during this peak month.

Figure 28

CMPS Dashboard - Percentage of Incident by Victim Age Groups



The visuals showing that adults were the common victims of an incident having a highest percentage among the age-group. It's most likely because adults are frequently presence in public spaces and active in workspace, which has a high possibility in risk environments. Even though adults have the highest average, there are still vulnerabilities in youths, seniors, and children. Given these insights will help the law enforcement and local government to identify which age-group based required more focused on awareness and community engagement to decrease victimization and enhance public safety.

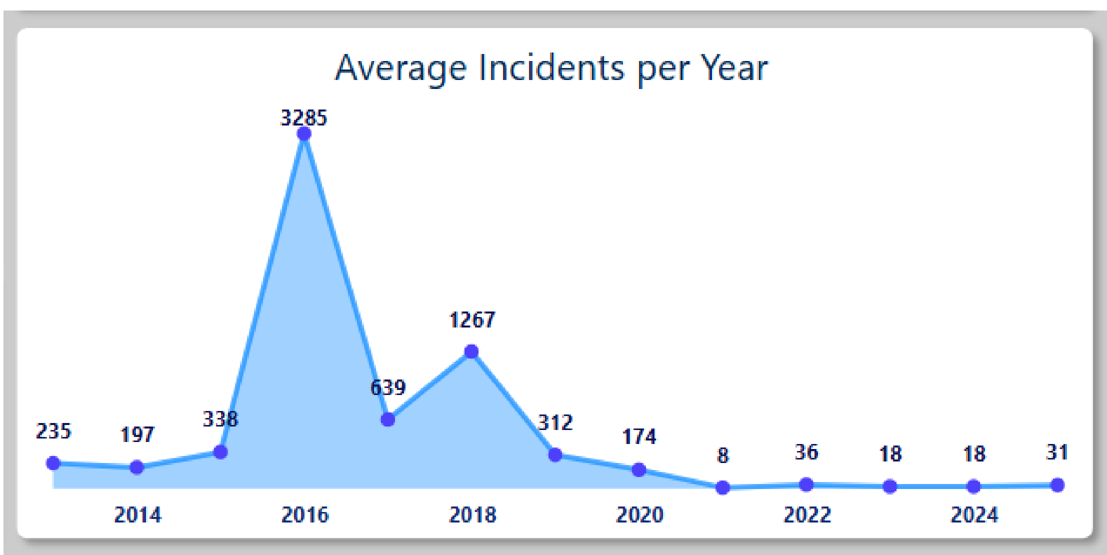
Figure 29

CMPS Dashboard - High Risk Time Period

In the figure it presents identifying what period the most frequent incidents happen. Evening, showed as the high-risk period which suggests incidents more likely to occur. These insights may be attributed to various factors, including alcohol-related offenses, social gatherings, and other factors that might happen during evening period. In line with this, local government and law enforcement could effectively deploy resources by increasing patrols and enhancing street lighting to ensure the safety of the community during peak hours.

Figure 30

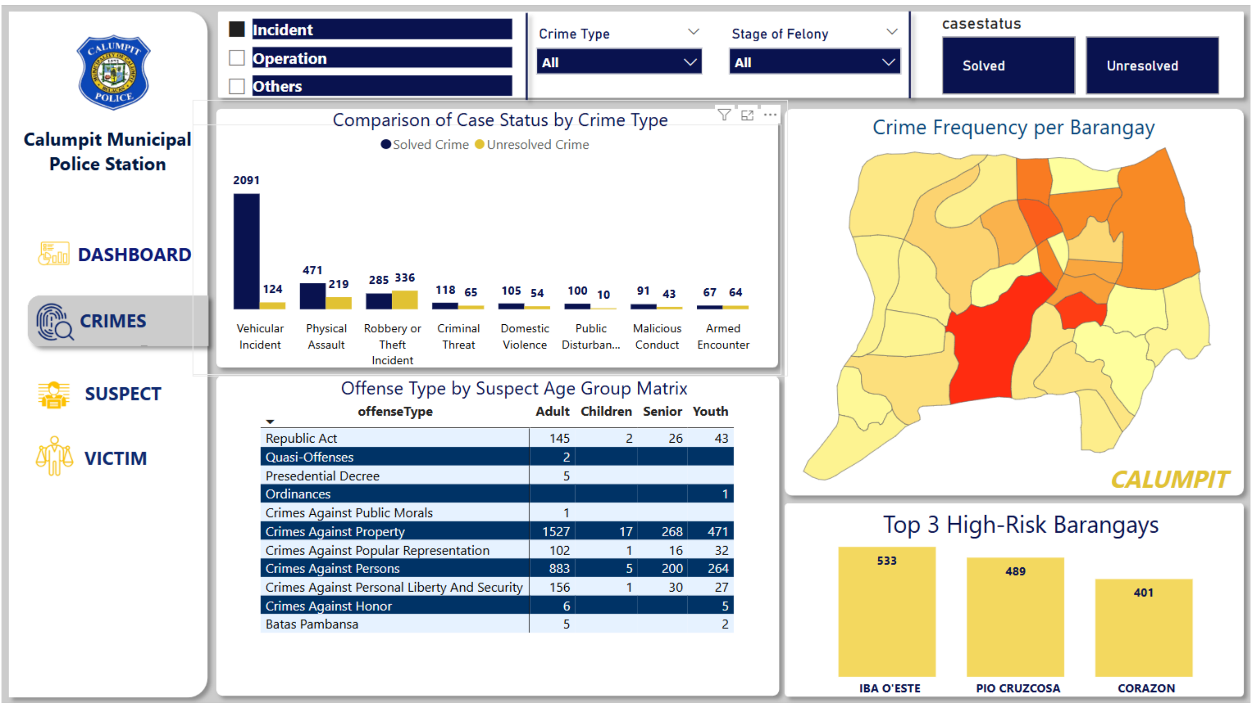
CMPS Dashboard - Average Incident Per Year



This Line Graph visual presents the highest average incidents per year from 2013-2025. The process used in Average Incident per Year is generated from the YearCommitted field and counting incidents from incidentTypeSpecific_Categorized to get the total average. During these years, it is revealed that in the year 2016 an increased number of incidents happen in the Municipality of Calumpit. There are factors that may cause why this year marks as peak year in the recorded datasets such as economic hardships or gaps in law enforcement coverage. Given these insights could help the local government to identify long term trends, further investigate to allocate police officers or barangay patrols more effectively which may result in decreasing the number of incidents in the future.

Figure 31

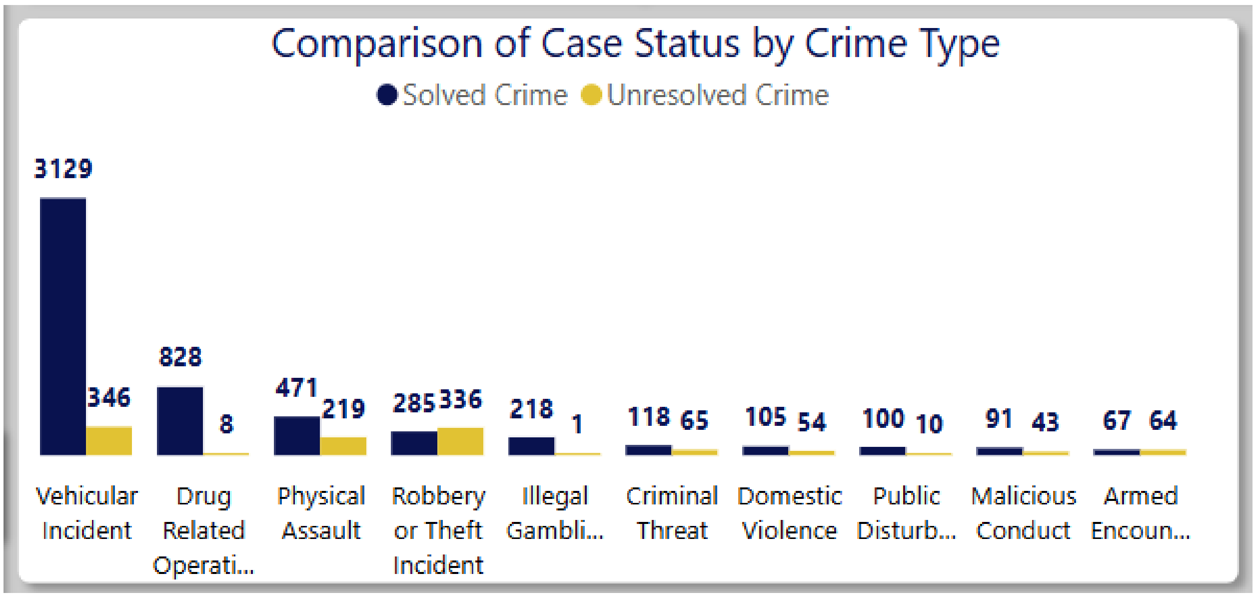
CMPS Crimes Tab



The crimes tab is a strategic component of the dashboard, offering an at-a-glance overview of criminal activity in Calumpit. The slicer for indicent and crime type allows users to isolate and analyze patterns for specific crimes, identify which barangays are more affected by incident type, and support scenario-based planning. The button slicer for case status is crucial for police effectiveness and encourages transparency and progress tracking in police case management. Lastly, the stage of felony adds legal depth to the crime assessment as it supports legal coordination and understanding of the nature of crimes per area of demographic. All in all, the visual clarity, interactivity, and drill-down capabilities empower law enforcement officers and investigators to monitor trends, measure case resolutions, and focus on crime hotspots.

Figure 32

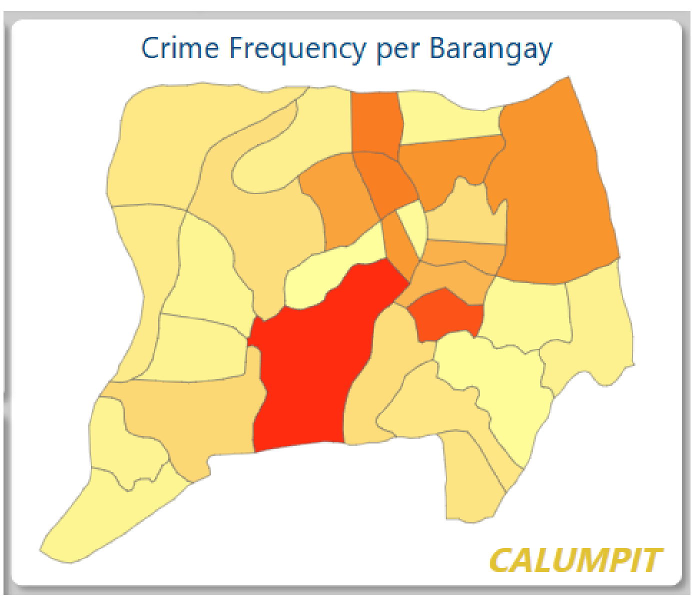
CMPS Crimes - Comparison of Case Status by Crime Type



Presenting the comparison of case status by crime type in Calumpit using bar chart, distinguishing between solved and unresolved cases. From the figure above, it reveals that Vehicular Incident have the highest solved cases with 3,129 and 346 unresolved cases indicating that it is the most common incident occurred in Calumpit. On the other hand, Drug-Related operation followed next with 828 solved cases, but surprisingly, only 8 remain for unresolved cases, this indicates an effective handling such cases. With these insights, it highlights which case needs further investigation for unresolved cases and maintain the solved cases.

Figure 33

CMPS Crimes - Crime Frequency per Barangay



This chart illustrates a shape map of Calumpit, Philippines, displaying crime frequency by barangay, labels for each barangay will appear on hover because displaying them permanently would overcrowd the map and reduce readability. The map has a color scale from yellow (low crime frequency) to dark red (high crime frequency). The darker the red, the higher the crime rate barangays are. This space visualization of crime data is vital in locating the high-crime areas, maximizing resource allocation, and guiding focused crime prevention efforts.

Figure 34

CMPS Crimes - Offense Type by Suspect Age Group Matrix

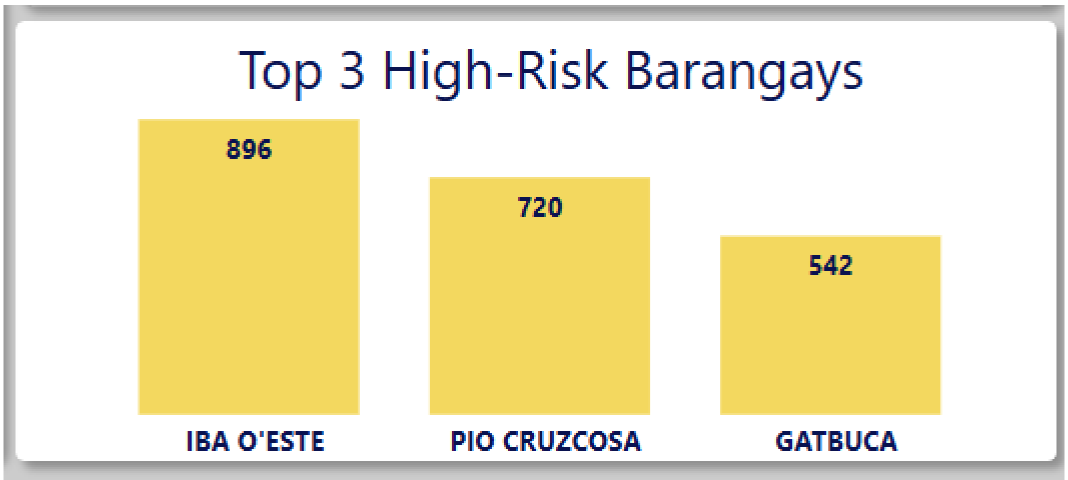
Offense Type by Suspect Age Group Matrix

offenseType	Adult	Children	Senior	Youth
Republic Act	882	8	181	270
Quasi-Offenses	2			
Presedential Decree	19	1	1	
Ordinances	3			3
Crimes Against Public Morals	1			
Crimes Against Public Interest	5			
Crimes Against Property	2140	20	375	584
Crimes Against Popular Representation	116	1	17	40
Crimes Against Persons	1079	8	226	303
Crimes Against Personal Liberty And Security	171	1	37	30
Crimes Against Honor	8			5
Crimes Against Chastity	7		3	2
Batas Pambansa	7			2

This table displays a matrix indicating the suspect's frequency of types of offense by age category in the CMPS Crimes. The rows are various categories of offense (Republic Act, Quasi-Offenses, Presidential Decree, Ordinances, and miscellaneous crimes against public morals, public interest, property, popular representation, persons, personal liberty and security, honor, and chastity, and Batas Pambansa). Columns are offense categories: Drug Offense, Violent Offense, Property Offense, etc., and age groups: Adult, Children, Senior, and Youth. Each cell contains the number of offenses in a specific offense category and age group. This analysis allows trends in criminal activity by offense type and age offending to be determined. The data allow more advanced understanding of crime patterns, which can result in more effective and targeted responses.

Figure 35

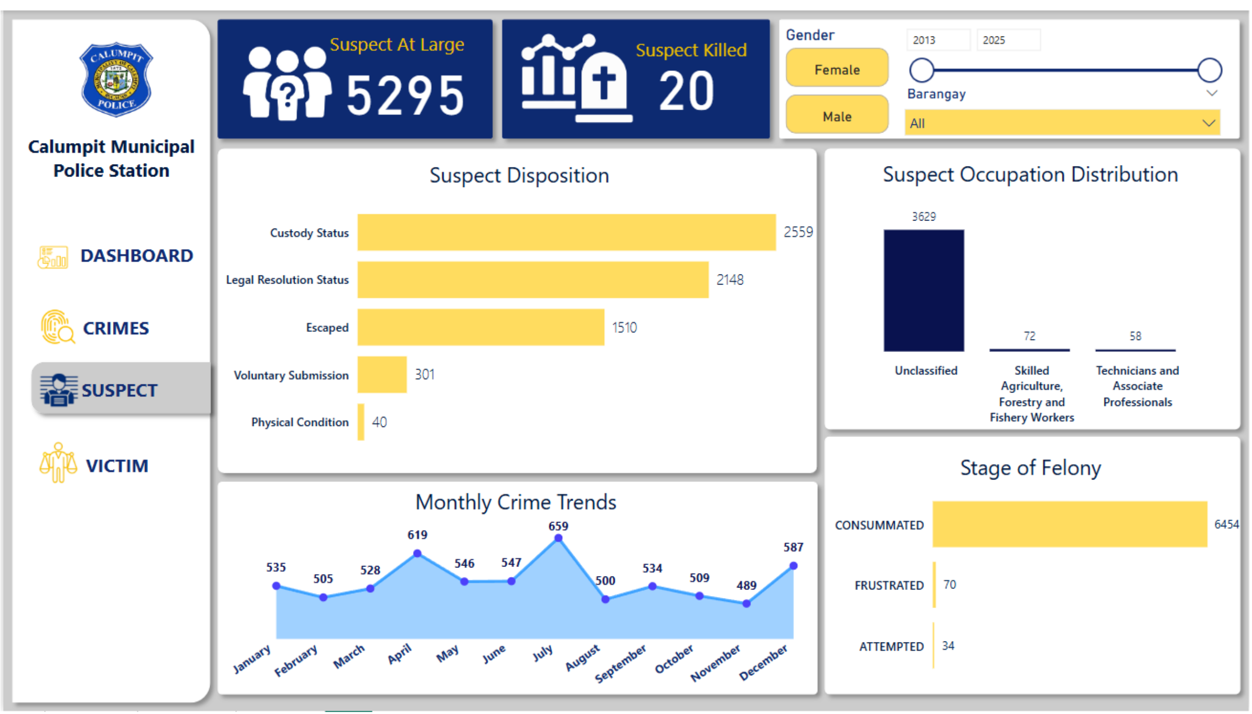
CMPS Crimes - Top 3 High-Risk Barangays



This bar chart presents the top 3 high-risk barangays in Calumpit based on the number of incidents recorded. The barangay of Iba O'Este showed with the highest rank for 2013-2025 record with 896 incidents, followed by Pio Cruzcosa with 720, and Gatbuca with 542. In line with this, Iba O'Este needs the most immediate attention and barangay patrols for risk mitigation among other barangays in Calumpit. Given this insightful visual will help the local government and barangay officials to develop an effective strategic plan for allocating patrols and prioritize high-risk with maintaining proactive measures in other barangay.

Figure 36

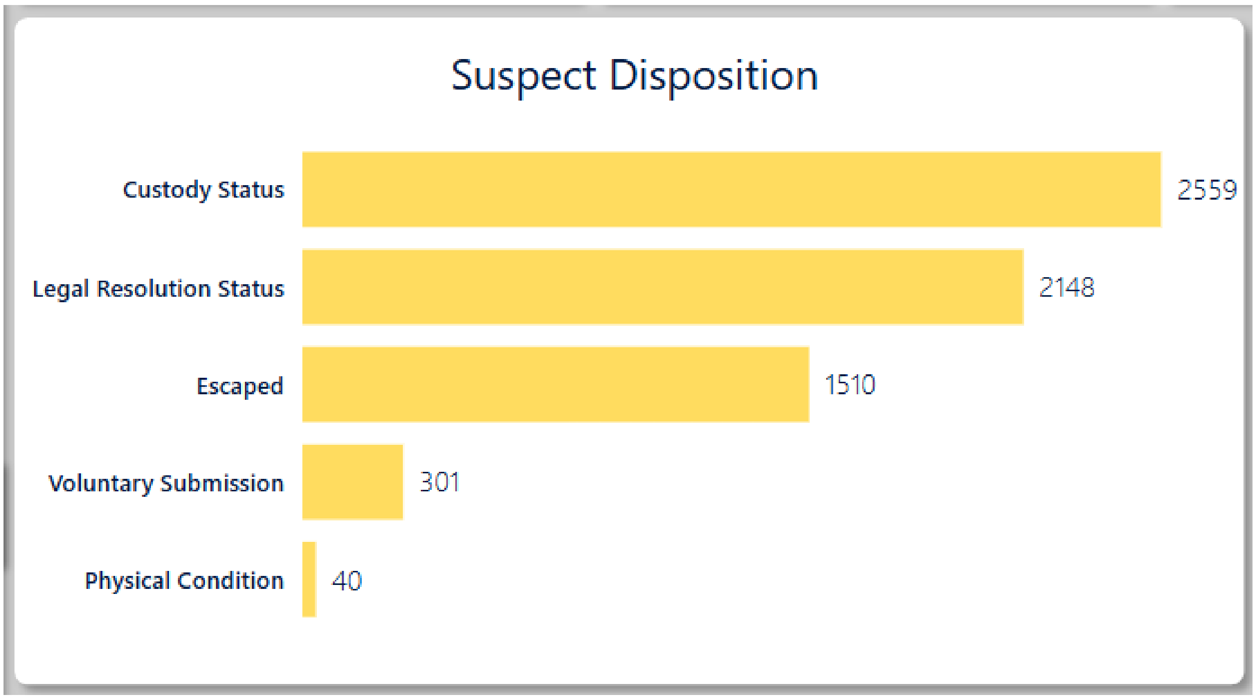
CMPS Suspect Tab



The CMPS dashboard's Suspect tab profiles Calumpit suspects through interactive filters (gender, barangay, date range) for focused analysis. Important indicators reveal suspects at large and killed, while charts present suspect disposition, occupation (largely unclassified), and the stage of felony (largely consummated). A monthly crime trend graph displays fluctuations, which help detect patterns and possible enforcement gaps. This tab facilitates data-driven crime prevention and resource planning.

Figure 37

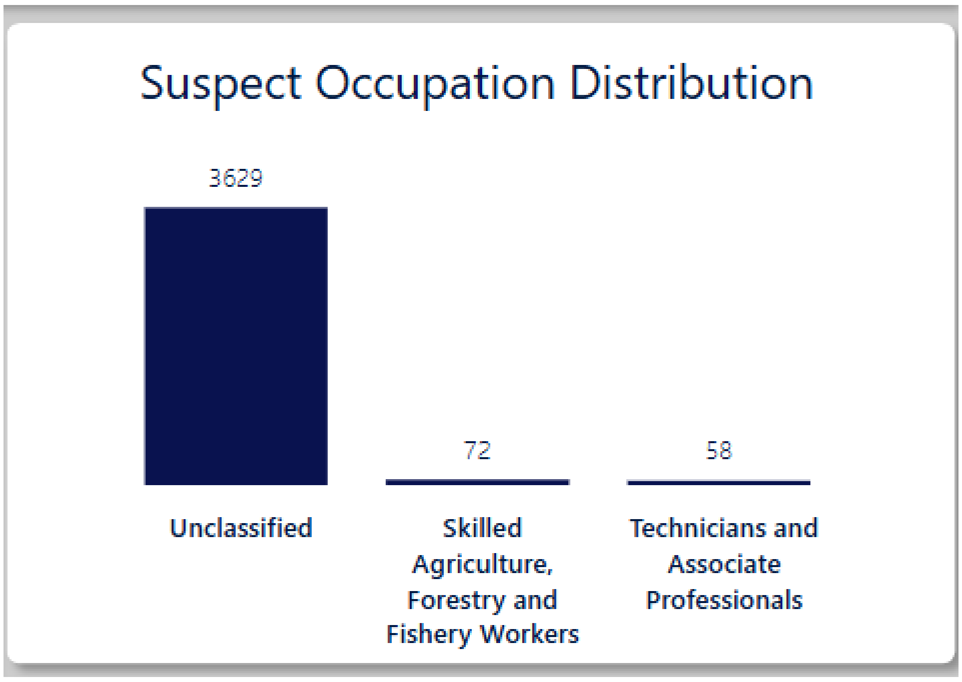
CMPS Suspect- Suspect Disposition



This bar chart illustrates the disposition of criminal incident suspects such as Custody Status (2559), Legal Resolution Status (2148), Escaped (1510), Voluntary Submission (301), and Physical Condition (40). The graph comfortably illustrates that the majority of suspects are in custody or have been brought to legal resolution, and a great percentage escaped. The comparatively small numbers for voluntary surrender and physical status suggest these are less frequent events. The graph here provides a quick snapshot of suspect statuses conducive to comparing law enforcement effectiveness and planning resources.

Figure 38

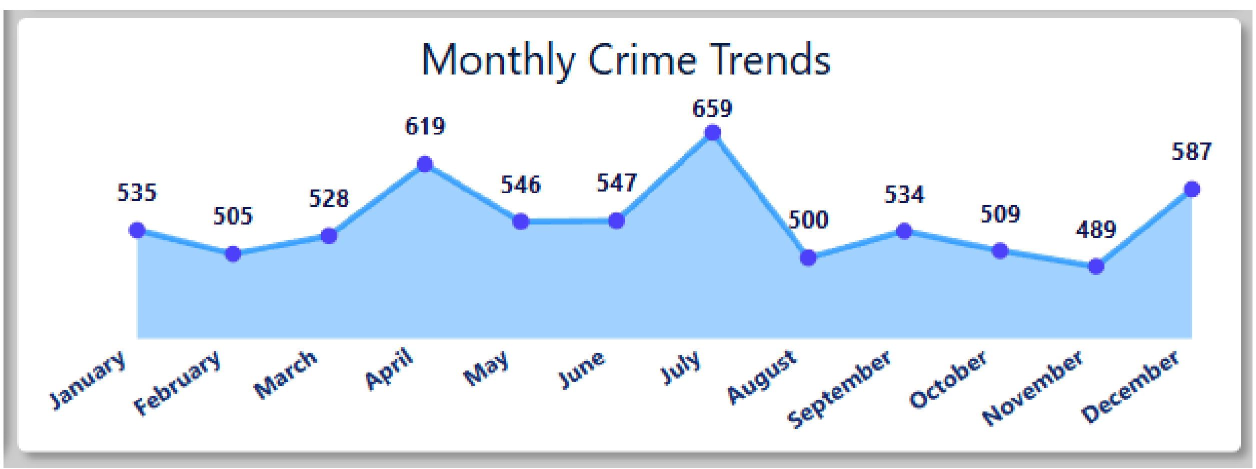
CMPS Suspect - Suspect Occupation Distribution



This bar graph shows the occupation distribution of suspects and indicates a broad spread. There is a vast majority (3629) that is "Unclassified" and relatively smaller quantities are accounted for in individual occupational categories such "Skilled Agriculture, Forestry and Fishery Workers," and "Technicians and Associate Professionals." This data reveals there is a need for improved collections of data for suspect occupations to enable more useful crime analysis and perhaps reveal occupational patterns of criminality.

Figure 39

CMPS Suspect - Monthly Crime Trends

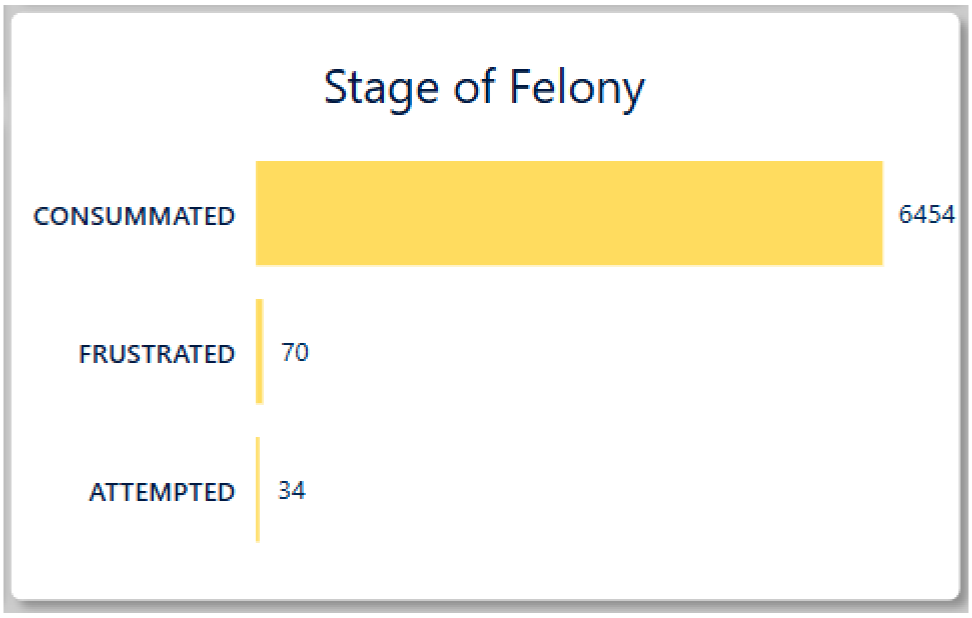


This monthly crime line graph is for crime trend analysis. The highest incidents of crime were in April and July, with 24 incidents for each of these months. In November, there was a dip to 18 incidents.

This graph presents a time-series view of various crime activity characters, which would be useful to potential discovery of seasonal patterns or patterns of times of the year that must be targeted by law enforcement.

Figure 40

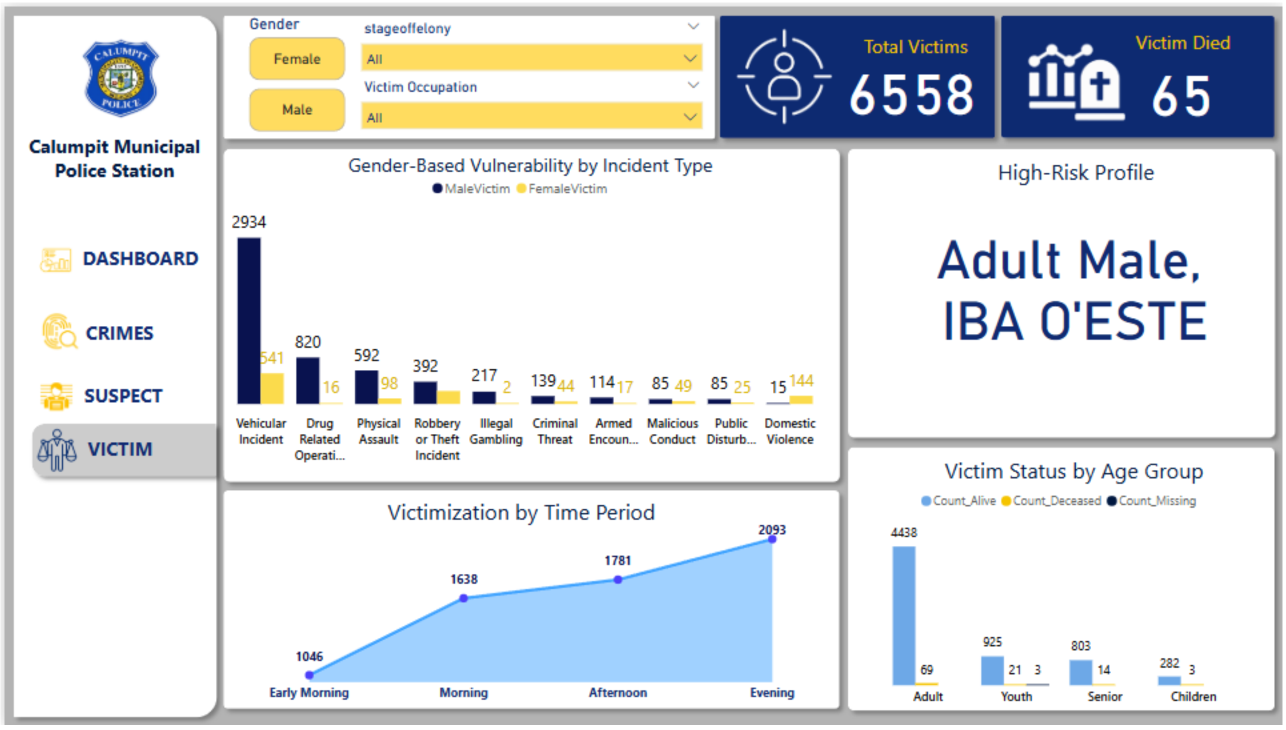
CMPS Suspect - Stage of Felony



This bar chart presents data on the stage of felony for criminal cases within the CMPS. This bar chart shows a large disparity in felony stages with most crimes (6454) consummated, far exceeding frustrated (70) and attempted (34) cases. This implies that the majority of crimes advance to their ultimate stages, showing that the crime-control measures need to be more proactive and in the early stages. The facts point to possible areas of compromise in crime deterrence and swift response.

Figure 41

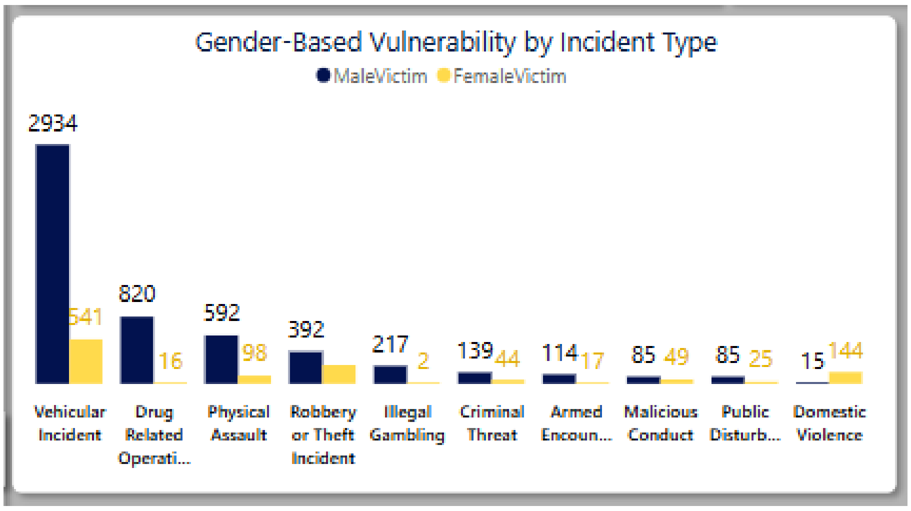
CMPS Victim Tab



The victim tab of the CMPS dashboard allows the users to shift their focus primarily on the information of victims that were involved in incidents. It features 3 slicers, for gender, stage of felony, and incident type. This interactivity enhances the analytical depth allowing for gender-specific vulnerability analysis, crime trend analysis by felony stage, and more customized filtering by incident type. Moreover, the two summary cards at the upper right of the figure offer at-a-glance key performance indicators (KPI) to provide context about the scale of victimization across Calumpit. These indicators allow the police to analyze number of victims against the fatality rate of each incidents. Particular use case scenario for this is if a user wants to focus on male victims involved in crimes at the "Attempted" stage to identify patterns and emerging threats. With its strong focus on victim-centered analytics, this tab supports the broader goal of improving community safety and victim support initiatives.

Figure 42

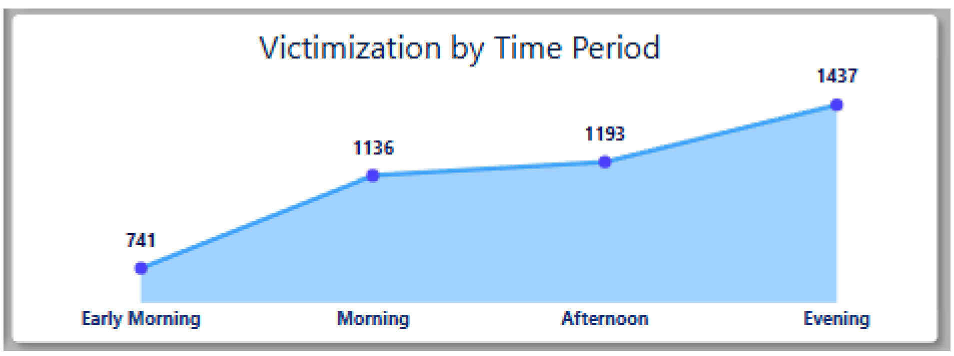
CMPS Victim Tab – Gender-Based Vulnerability by Incident Type



This clustered column chart compares the number of male vs female victims across all the values of incident type. This visualization helps in identifying which demographics are most vulnerable to certain crimes. In the figure above, we can see that in almost all, except for one, incident types present, male victims significantly outnumber female victims. This suggests that male victims may have higher exposure to public or street-related threats. The prominence of female victims in domestic violence cases can guide community outreach, legal assistance, or protective services targeting domestic abuse.

Figure 43

CMPS Victim Tab – Victimization by Time Period



This line graph shows the amount of victim cases or victimizations at various periods of the day. As we can see from the figure, it is an uptrend where early morning has the fewest (741) recorded victim cases with a sharp gradual increase up to evening having the highest (1437) percentage of recorded victim cases across all time periods. Respectively, the morning (1136) and afternoon (1193) fall in the center. The timing of police patrols and crime prevention tactics depend heavily on this information. The station can better manage resources and possibly lower victimization rates by detecting high-risk times.

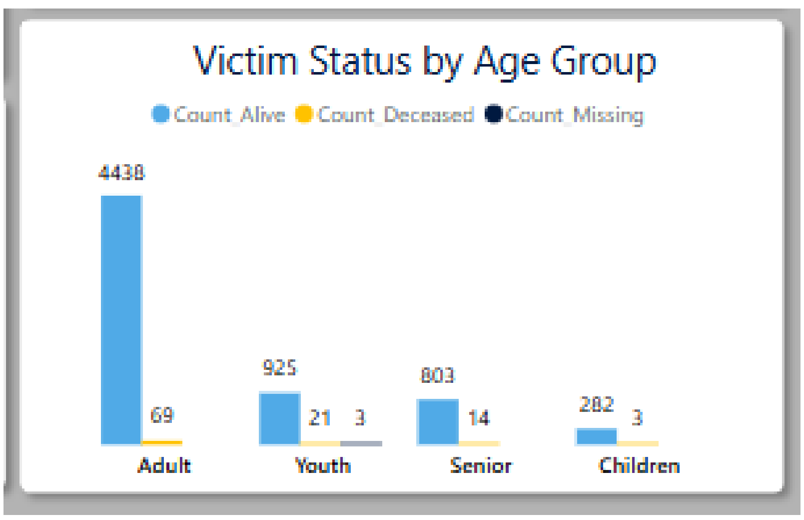
Figure 44

CMPS Victim Tab – High-Risk Profile

This card synthesizes the most common victim profile based on the dataset and the slicers applied. The figure shows the Victim Age Group (Adult), Victim Gender (Male), and the Barangay (Iba O'Este) showing the most vulnerable demographic and their common geographic location. Knowing this information would allow the police to tailor intervention programs, targeted awareness or safety campaigns by communicating and collaborating with barangay officials as well as strengthening visibility and presence in Iba O'Este.

Figure 45

CMPS Victim Tab – Victim Status by Age Group



The clustered column chart breaks down the status of victims across four key age groups, offering a more victim-centered analysis vital in understanding who the victims are, not just by count but by impact severity per age group. From the figure above, Adults (aged 25 - 64) form the largest group of victims with 4,438 survivors and 69 (1.53% fatality rate for adults) fatalities. This aligns with Figure 25, suggesting that adults have higher exposure to incidents due to mobility, employment and active presence in public spaces. This suggests that the Adults are not necessarily the most vulnerable, but are statistically at higher risk. Youth victims (aged 15-24) had 925 survivors, 21 fatalities (2.21% fatality rate for youths), and 3 missing cases. Being the second highest age group for victimization shows strong vulnerability rate compared to Adults. Youth may be susceptible to drug-related operations, gang activity, or other incidents which can escalate into fatal scenarios. The presence of missing youth cases also introduces a critical concern about abduction, exploitation, or trafficking, instigating immediate investigation and intervention. Moving forward, senior citizens (aged 65 and over) form a modest portion of the dataset having 803 survivors and 14 fatalities (1.7% fatality rate in senior citizens). Seniors may be more vulnerable due to physical frailty, having slower response time or being targeted for theft or abuse. Lastly, while having the smallest group by count, Children victims (aged 0-14) had 282 survivors and 3 fatalities (1.05% fatality rate). Their inability to protect themselves poses a significant threat to their age groups. Intensive safeguarding efforts and the need for collaboration with child welfare agencies, schools and barangay level child protection units are needed.

Conclusion

Overall, the project successfully implements descriptive analytics by creating an interactive dashboard that will present an overview insight of incidents that happened from 2013-2025 in Calumpit, Bulacan. The Calumpit Municipal Police Station (CMPS) will benefit from the project, as well as the local government and barangay officials by understanding the most common incident that happens in the area. The Dashboard has 4-page features, including the dashboard page as the first page presenting the overview insights, followed by crimes page showing the drill down analysis of barangays incidents with heatmaps visuals for identifying which barangays has the frequent number of incidents then, the suspect and victim page highlighting the demographics insights. Ultimately, this visualization project will help the local government of Calumpit, Bulacan to effectively allocate law enforcement in high-risk areas to improve crime prevention and decrease the number of incidents in the future and for more secure and safety community.